14 Partial Derivatives

14.1 Functions of Several Variables

Functions of Two Variables

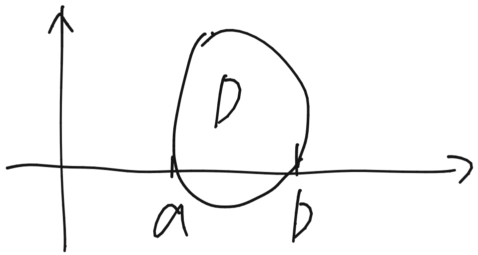
Functions of Two Variables (1 of 2)

The temperature T at a point on the surface of the earth at any given time depends on the longitude x and latitude y of the point. We can think of T as being a function of the two variables x and y, or as a function of the pair (x, y). We indicate this functional dependence by writing $T = f(x, y)$.

The volume V of a circular cylinder depends on its radius r and its height h. In fact, we know that $V = \pi r^2 h$. We say that V is a function of r and h, and we can write $v(r, h) = \pi r^2 h$.

Functions of Two Variables (2 of 2)

Definition **A function *f* of two variables** is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the **domain** of f and its **range** is the set of values that f takes on, that is, $\{f(x,y) \mid (x,y) \in D\}$.



$D \subset \mathbb{R}^2$

$[a, b] \subset \mathbb{R}$

We often write $z = f(x, y)$ to make explicit the value taken on by f at the general point (x, y). The variables x and y are **independent variables** and z is the **dependent variable**. [Compare this with the notation $y = f(x)$ for functions of a single variable.]

EXAMPLE 1.1 Finding the Domain of a Function of Two Variables

Find and sketch the domain for (a) $f(x, y) = x \ln y$ and (b) $g(x, y) = \dfrac{2x}{y - x^2}$.

$D_f = \{(x, y) \in R^2 \mid y > 0\}$

$D_g = \{(x, y) \in R^2 \mid y \neq x^2\}$

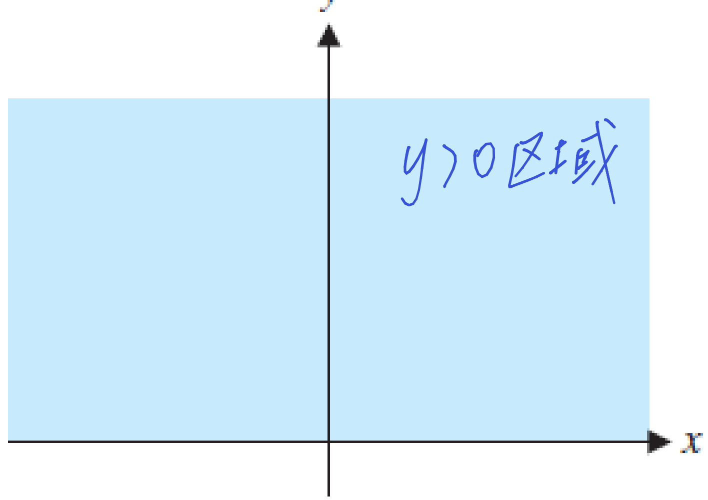


FIGURE 12.1a
The domain of $f(x, y) = x \ln y$

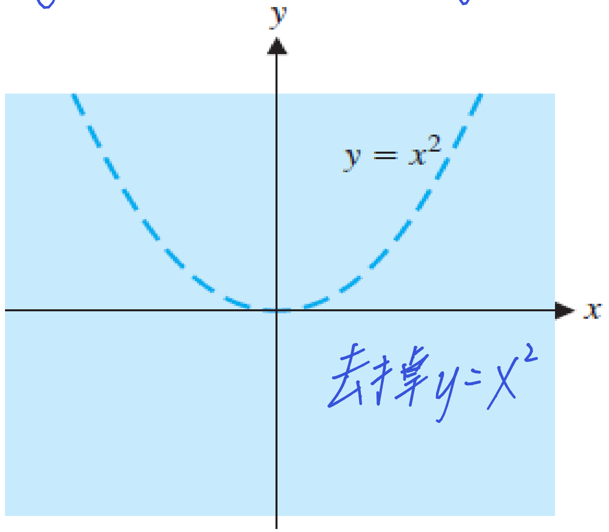


FIGURE 12.1b
The domain of $g(x, y) = \dfrac{2x}{y - x^2}$

Example 4 (1 of 6)

In 1928 Charles Cobb and Paul Douglas published a study in which they modeled the growth of the American economy during the period 1899−1922.

They considered a simplified view of the economy in which production output is determined by the amount of labor involved and the amount of capital invested.

While many other factors affect economic performance, their model proved to be remarkably accurate.

Example 4 (2 of 6)

function Cobb and Douglas used to model production was of the form

$$P(L, K) = bL^{\alpha}K^{1-\alpha} \quad \textbf{1}$$

柯布-道格拉斯生产函数

where *P* is the total production (the monetary value of all goods produced in a year), *L* is the amount of labor (the total number of person-hours worked in a year), and *K* is the amount of capital invested (the monetary worth of all machinery, equipment, and buildings).

Example 4 (3 of 6)

Its domain is $\{(L,k) \mid L \geq 0, k \geq 0\}$ because *L* and *K* represent labor and capital and are therefore never negative.

劳动投入 资本投入

Example 4 (4 of 6)

Cobb and Douglas used economic data published by the government to obtain Table 2.

Year	*P*	*L*	*K*
1899	100	100	100
1900	101	105	107
1901	112	110	114
1902	122	117	122
1903	124	122	131
1904	122	121	138
1905	143	125	149
1906	152	134	163
1907	151	140	176
1908	126	123	185
1909	155	143	198
1910	159	147	208

Year	*P*	*L*	*K*
1911	153	148	216
1912	177	155	226
1913	184	156	236
1914	169	152	244
1915	189	156	266
1916	225	183	298
1917	227	198	335
1918	223	201	366
1919	218	196	387
1920	231	194	407
1921	179	146	417
1922	240	161	431

Table 2

Example 4 (5 of 6)

They took the year 1899 as a baseline and *P*, *L*, and *K* for 1899 were each assigned the value 100. The values for other years were expressed as percentages of the 1899 values.

Cobb and Douglas used the method of least squares to fit the data of Table 2 to the function

2 $$P(L,K) = 1.01L^{0.75}K^{0.25}$$

Example 4 (6 of 6)

If we use the model given by the function in Equation 2 to compute the production in the years 1910 and 1920, we get the values

$$P(147,\ 208) = 1.01(147)^{0.75}(208)^{0.25} \approx 161.9$$

$$P(194,\ 407) = 1.01(194)^{0.75}(407)^{0.25} \approx 235.8$$

which are quite close to the actual values, 159 and 231.

The production function (1) has subsequently been used in many settings, ranging from individual firms to global economics. It has become known as the **Cobb-Douglas production function**.

Graphs (1 of 3)

Another way of visualizing the behavior of a function of two variables is to consider its graph.

Definition If f is a function of two variables with domain D, then the **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and (x, y) is in D.

The graph of a function f of two variables is a surface S with equation $z = f(x, y)$.

Graphs (2 of 3)

We can visualize the graph *S* of *f* as lying directly above or below its domain *D* in the *xy*-plane (see Figure 5).

二元函数 $f(x,y)$
图形为三维空间的一个曲面
z值随x, y变化而变化

x, y 横纵坐标
$z=f(x,y)$ 为垂直高度

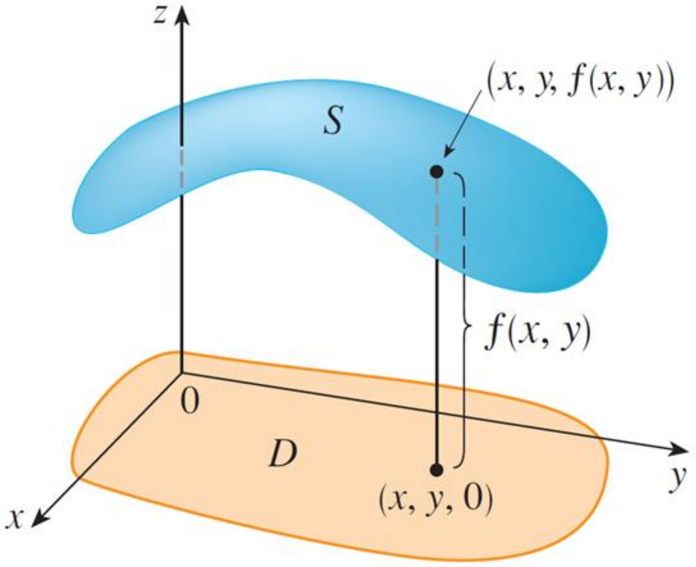


Figure 5

Graphs (3 of 3)

The function $f(x, y) = ax + by + c$ is called as a **linear function**.

The graph of such a function has the equation R^3 线性函数的图形为平面

$$z = ax + by + c \quad \text{or} \quad ax + by - z + c = 0$$

so it is a plane. In much the same way that linear functions of one variable are important in single-variable calculus, we will see that linear functions of two variables play a central role in multivariable calculus.

Example 6

Sketch the graph of $g(x,y) = \sqrt{9 - x^2 - y^2}$.

R^3 二次函数 → 球面

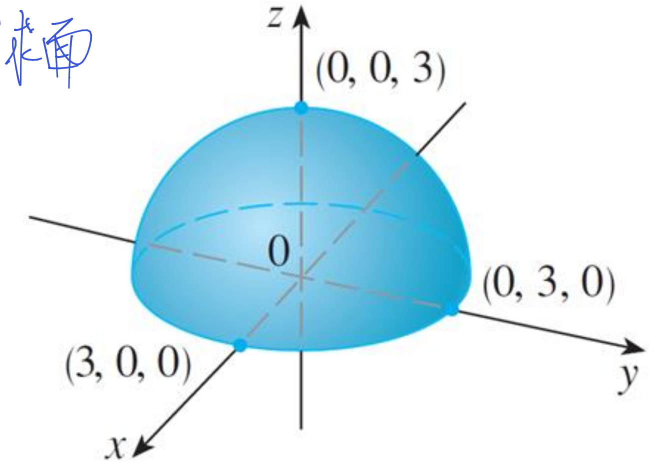
Solution:

In Example 2 we found that the domain of g is the disk with center (0, 0) and radius 3. The graph of t has equation $z = \sqrt{9 - x^2 - y^2}$.

球面

We square both sides of this equation to obtain $z^2 = 9 - x^2 - y^2$, or $x^2 + y^2 + z^2 = 9$, which we recognize as an equation sphere with center the origin and radius 3.

But, since $z \geq 0$, the graph of g is just the top half of this sphere (see Figure 7).



Graph of $g(x,y) = \sqrt{9 - x^2 - y^2}$

Figure 7

Functions of Three or More Variables

Functions of Three or More Variables (1 of 5)

A **function of three variables**, f, is a rule that assigns to each ordered triple (x, y, z) in a domain $D \subset \mathbb{R}^3$ a unique real number denoted by $f(x, y, z)$.

For instance, the temperature T at a point on the surface of the earth depends on the longitude x and latitude y of the point and on the time t, so we could write $T = f(x, y, t)$.

Example 14

Find the domain of *f* if

$$f(x,\ y,\ z) = \ln(z - y) + xy \sin z$$

Solution:

The expression for $f(x, y, z)$ is defined as long as $z - y > 0$, so the domain of *f* is

$$D = \{(x,\ y,\ z) \in \mathbb{R}^3 \mid z < y\}$$

This is a **half-space** consisting of all points that lie above the plane $z = y$.

14.2 Limits and Continuity

Limits of Functions of Two Variables

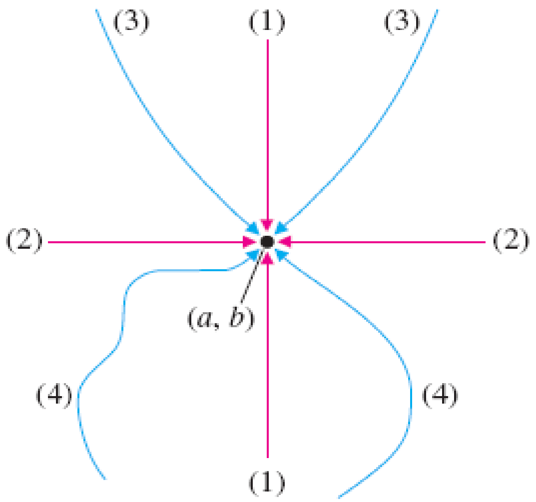
❑Limits

Recall: for a function of a single variable

$$\lim_{x \to a} f(x) = L$$

For a function of two variables

$$\lim_{(x,y) \to (a,b)} f(x, y) = L,$$



多变量极限→高度

FIGURE 12.15
Various paths to (a, b)

Limits of Functions of Two Variables (1 of 7)

Let's compare the behavior of the functions

$$f(x,y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$$

and

$$g(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$$

as *x* and *y* both approach 0 [and therefore the point (*x*, *y*) approaches the origin]. Note the domain of the function is R^2 excluding the origin

Tables 1 and 2 show values of $f(x, y)$ and $g(x, y)$, correct to three decimal places, for points (x, y) near the origin. (Notice that neither function is defined at the origin.)

$x \backslash y$	−1.0	−0.5	−0.2	0	0.2	0.5	1.0
−1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455
−0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
−0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0	0.841	0.990	1.000		1.000	0.990	0.841
0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455

Values of $f(x, y)$

Table 1

$x \backslash y$	-1.0	-0.5	-0.2	0	0.2	0.5	1.0
-1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000
-0.5	-0.600	0.000	0.724	1.000	0.724	0.000	-0.600
-0.2	-0.923	-0.724	0.000	1.000	0.000	-0.724	-0.923
0	-1.000	-1.000	-1.000		-1.000	-1.000	-1.000
0.2	-0.923	-0.724	0.000	1.000	0.000	-0.724	-0.923
0.5	-0.600	0.000	0.724	1.000	0.724	0.000	-0.600
1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000

Values of $g(x, y)$

Table 2

Limits of Functions of Two Variables (4 of 7)

It appears that as (x, y) approaches $(0, 0)$, the values of $f(x, y)$ are approaching 1 whereas the values of $g(x, y)$ aren't approaching any particular number. It turns out that these guesses based on numerical evidence are correct, and we write

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2} = 1 \qquad \text{and}$$

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} \qquad \text{does not exist}$$

Limits of Functions of Two Variables (5 of 7)

In general, we use the notation

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

to indicate that the values of $f(x, y)$ approach the number L as the point (x, y) approaches the point (a, b) (staying within the domain of f).

Limits of Functions of Two Variables (6 of 7)

In other words, we can make the values of $f(x, y)$ as close to L as we like by taking the point (x, y) sufficiently close to the point (a, b), but not equal to (a, b). A more precise definition follows.

1 Definition Let f be a function of two variables whose domain D includes points arbitrarily close to (a, b). Then we say that the **limit of *f*(*x*, *y*) as** (x, y) **approaches** (***a*, *b***) is L and we write

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

if for every number $\varepsilon < 0$ there is a corresponding number $\delta > 0$ such that

$$\text{if} \quad (x,y) \in D \quad \text{and} \quad 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \quad \text{then} \quad |f(x,y) - L| < \varepsilon$$

Limits of Functions of Two Variables (7 of 7)

Other notations for the limit in Definition 1 are

$$\lim_{\substack{x \to a \\ y \to b}} f(x, y) = L \qquad \text{and} \qquad f(x, y) \to L \text{ as } (x, y) \to (a, b)$$

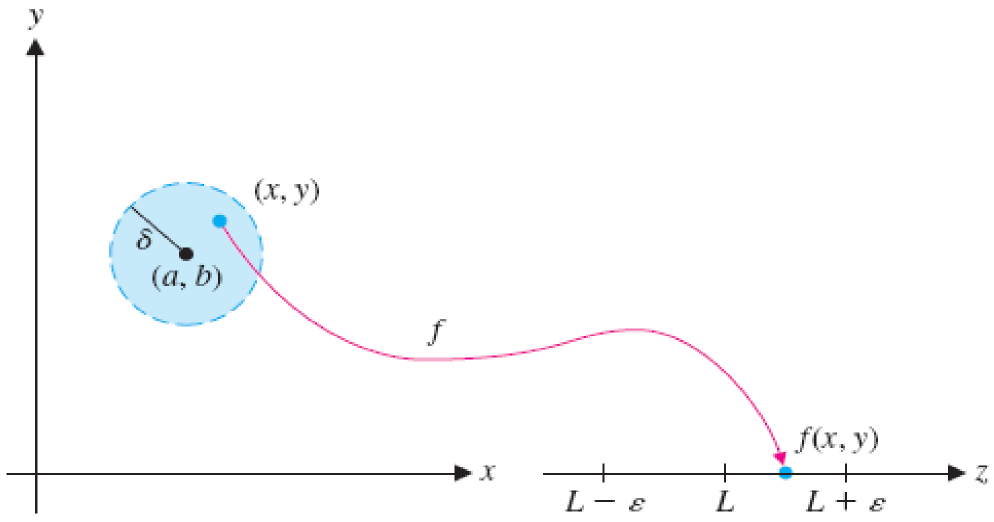


FIGURE 12.14
The definition of limit

Showing That a Limit Does Not Exist

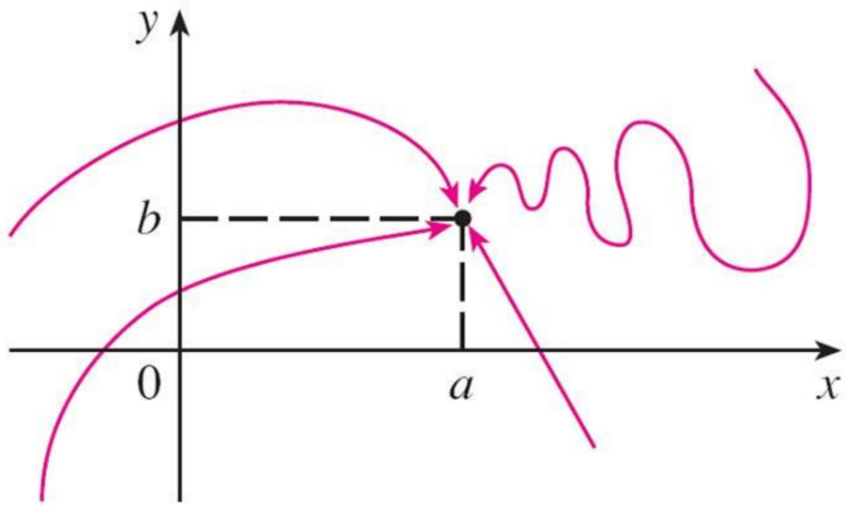
Showing That a Limit Does Not Exist (1 of 4)

For functions of a single variable, when we let *x* approach *a*, there are only two possible directions of approach, from the left or from the right.

We know that if $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$, then $\lim_{x \to a} f(x)$ does not exist.

Showing That a Limit Does Not Exist (2 of 4)

For functions of two variables the situation is not as simple because we can let (x, y) approach (a, b) from an infinite number of directions in any manner whatsoever (see Figure 3) as long as (x, y) stays within the domain of f.



Different paths approaching (a, b)

Figure 3

Showing That a Limit Does Not Exist (3 of 4)

Definition 1 says that the distance between $f(x, y)$ and L can be made arbitrarily small by making the distance from (x, y) to (a, b) sufficiently small (but not 0).

The definition refers only to the *distance* between (x, y) and (a, b). It does not refer to the direction of approach.

Therefore, if the limit exists, then $f(x, y)$ must approach the same limit *no matter how* (x, y) approaches (a, b).

Showing That a Limit Does Not Exist (4 of 4)

Thus one way to show that $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist is to find different paths of approach along which the function has different limits.

If $f(x,y) \to L_1$ as $(x, y) \to (a, b)$ along a path C_1 and $f(x, y) \to L_2$ as $(x, y) \to (a, b)$ along a path C_2, where $L_1 \neq L_2$, then $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist.

Example 1

Show that $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2-y^2}{x^2+y^2}$ does not exist.

Solution:

Let $f(x,y)=\dfrac{x^2-y^2}{x^2+y^2}$.

First let's approach (0, 0) along the x-axis. On this path $y = 0$ for every point (x, y), so the function becomes $f(x,0)=\dfrac{x^2}{x^2}=1$ for all $x\neq 0$ and thus

$$f(x,\ y)\to 1 \qquad \text{as} \qquad (x,\ y)\to(0,0) \text{ along the } x-axis$$

Example 1 – Solution (1 of 2)

We now approach along the y-axis by putting $x = 0$.

Then $f(0, y) = \dfrac{-y^2}{y^2} = -1$ for all $y \neq 0$, so

$$f(x, y) \to -1 \quad \text{as} \quad (x, y) \to (0, 0) \text{ along the } y-\text{axis}$$

(See Figure 4.)

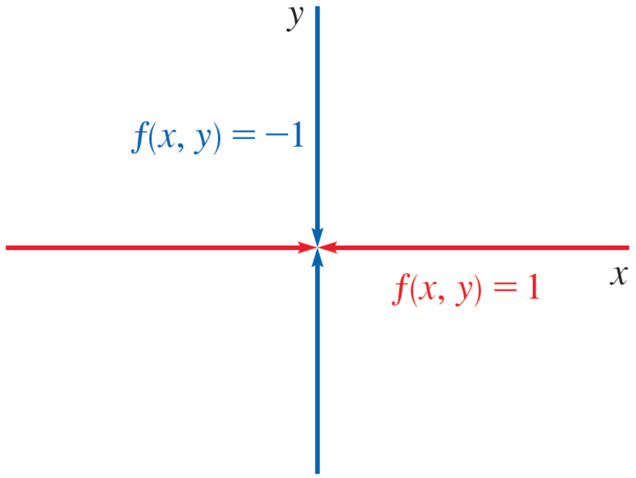


Figure 4

路径的选择

1. $y = k$
2. $y = kx$
3. $y = kx^2$
4. $y = x^n$

Example 1 – Solution (2 of 2)

Since f has two different limits as (x, y) approaches (0, 0) along two different lines, the given limit does not exist. (This confirms the conjecture we made on the basis of numerical evidence at the beginning of this section.)

Properties of Limits

Properties of Limits (1 of 5)

Just as for functions of one variable, the calculation of limits for functions of two variables can be greatly simplified by the use of properties of limits.

The Limit Laws can be extended to functions of two variables. Assuming that the indicated limits exist, we can state these laws verbally as follows:

Sum Law	**1.** The limit of a sum is the sum of the limits.
Difference Law	**2.** The limit of a difference is the difference of the limits.
Constant Multiple Law	**3.** The limit of a constant times a function is the constant times the limit of the function.

Properties of Limits (2 of 5)

Product Law **4.** The limit of a product is the product of the limits.

Quotient Law **5.** The limit of a quotient is the quotient of the limits (provided that the limit of the denominator is not 0).

You are asked to prove the following special limits:

2 $\lim_{(x,y)\to(a,b)} x = a$ $\qquad \lim_{(x,y)\to(a,b)} y = b$ $\qquad \lim_{(x,y)\to(a,b)} c = c$

Properties of Limits (3 of 5)

A polynomial function of two variables (or polynomial, for short) is a sum of terms of the form $cx^m y^n$, where c is a constant and m and n are nonnegative integers. A **rational function** is a ratio of two polynomials. For instance,

$$p(x, y) = x^4 + 5x^3y^2 + 6xy^4 - 7y + 6$$

is a polynomial, whereas

$$q(x, y) = \frac{2xy + 1}{x^2 + y^2}$$

is a rational function.

Properties of Limits (4 of 5)

The special limits in (2) along with the limit laws allow us to evaluate the limit of any polynomial function p by direct substitution:

有理函数, 多项式函数

极限直接代入

3 $$\lim_{(x,y)\to(a,b)} p(x,y) = p(a,b)$$

Similarly, for any rational function $q(x, y) = p(x, y)\,/r(x, y)$ we have

4 $$\lim_{(x,y)\to(a,b)} q(x,y) = \lim_{(x,y)\to(a,b)} \frac{p(x,y)}{r(x,y)} = \frac{p(a,b)}{r(a,b)} = q(a,b)$$

provided that (a, b) is in the domain of q.

Example 4

Evaluate $\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y)$.

Solution:

Let $f(x,y) = x^2y^3 - x^3y^2 + 3x + 2y$ is a polynomial, we can find the limit by direct substitution:

$$\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y) = 1^2 \cdot 2^3 - 1^3 \cdot 2^2 + 3 \cdot 1 + 2 \cdot 2 = 11$$

Properties of Limits (5 of 5)

The Squeeze Theorem also holds for functions of two or more variables.

$$g(x, y) \leq f(x, y) \leq h(x, y)$$

$$\lim_{(x,y)\to(a,b)} g(x, y) = L \text{ and } \lim_{(x,y)\to(a,b)} h(x, y) = L$$

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

夹逼定理
1.适用于证明极限存在
不适用于证明不存在
2.多变量中
结合极坐标形式
尤其函数有辐射
对称性时

Continuity

Continuity (1 of 4)

We know that evaluating limits of *continuous* functions of a single variable is easy.

It can be accomplished by direct substitution because the defining property of a continuous function is $\lim_{x \to a} f(x) = f(a)$.

Continuous functions of two variables are also defined by the direct substitution property.

6 Definition A function *f* of two variables is called **continuous at** (*a*, *b*) if

$$\text{连续} \rightarrow \lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

条件
极限存在（该点）
函数值存在（该点）
极限值=函数值（该点）

We say *f* is **continuous on** *D* if *f* is continuous at every point (*a*, *b*) in *D*.

Continuity (2 of 4)

The intuitive meaning of continuity is that if the point (x, y) changes by a small amount, then the value of $f(x, y)$ changes by a small amount.

This means that a surface that is the graph of a continuous function has no hole or break.

We have already seen that limits of polynomial functions can be evaluated by direct substitution. It follows by the definition of continuity that *all polynomials are continuous on* $\mathbb{R}^2$.

Continuity (3 of 4)

Likewise, Equation 4 shows that *any rational function is continuous on its domain*.

In general, using properties of limits, you can see that sums, differences, products, and quotients of continuous functions are continuous on their domains.

Example 7

Where is the function $f(x,y)=\dfrac{x^2-y^2}{x^2+y^2}$ continuous?

Solution:

The function f is discontinuous at (0, 0) because it is not defined there.

Since f is a rational function, it is continuous on its domain, which is the set

$D = \{(x, y) \mid (x, y) \neq (0, 0)\}$.

Continuity (4 of 4)

Just as for functions of one variable, composition is another way of combining two continuous functions to get a third.

In fact, it can be shown that if f is a continuous function of two variables and g is a continuous function of a single variable that is defined on the range of f, then the composite function $h = g \circ f$ defined by $h(x, y) = g(f(x, y))$ is also a continuous function.

Functions of Three or More Variables

Functions of Three or More Variables (1 of 4)

Everything that we have done in this section can be extended to functions of three or more variables.

The notation

$$\lim_{(x,y,z)\to(a,b,c)} f(x,y,z) = L$$

means that the values of $f(x, y, z)$ approach the number L as the point (x, y, z) approaches the point (a, b, c) (staying within the domain of f).

Functions of Three or More Variables (2 of 4)

Because the distance between two points (x, y, z) and (a, b, c) in $\mathbb{R}^3$ is given by $\sqrt{(x-a)^2+(y-b)^2+(z-c)^2}$, we can write the precise definition as follows: for every number $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

if (x, y, z) is in the domain of f and

$$0 < \sqrt{(x-a)^2+(y-b)^2+(z-c)^2} < \delta$$

$$\text{then } |f(x,y,z)-L| < \varepsilon$$

Functions of Three or More Variables (3 of 4)

The function *f* is **continuous** at (*a*, *b*, *c*) if

$$\lim_{(x,y,z)\to(a,b,c)} f(x, y, z) = f(a, b, c)$$

For instance, the function

$$f(x, y, z) = \frac{1}{x^2 + y^2 + z^2 - 1}$$

is a rational function of three variables and so is continuous at every point in $\mathbb{R}^3$ except where $x^2 + y^2 + z^2 = 1$. In other words, it is discontinuous on the sphere with center the origin and radius 1.

Functions of Three or More Variables (4 of 4)

We can write the definitions of a limit for functions of two or three variables in a single compact form as follows.

7 If f is defined on a subset D of $\mathbb{R}^n$, then $\lim_{x \to a} f(x) = L$ means that for every number $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$\text{if} \quad \mathbf{x} \in D \quad \text{and} \quad 0 < |\mathbf{x} - \mathbf{a}| < \delta \quad \text{then} \quad |f(\mathbf{x}) - L| < \varepsilon$$

14.3 Partial Derivatives

偏导数

Recall that for a function f of a single variable, we define the derivative function as

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

單變量

Partial Derivatives of Functions of Two Variables

多变量

Partial Derivatives of Functions of Two Variables (1 of 14)

On a hot day, extreme humidity makes us think the temperature is higher than it really is, whereas in very dry air we perceive the temperature to be lower than the thermometer indicates.

The National Weather Service has devised the *heat index* (also called the temperature-humidity index, or humidex, in some countries) to describe the combined effects of temperature and humidity.

The heat index I is the perceived air temperature when the actual temperature is T and the relative humidity is H.

So I is a function of T and H and we can write $I = f(T, H)$.

The following table of values of I is an excerpt from a table compiled by the National Weather Service.

Relative humidity (%)

Actual temperature (°F) T \ H	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	125	130	135	141	146
98	114	118	123	127	133	138	144	150	157
100	119	124	129	135	141	147	154	161	168

Heat index I as a function of temperature and humidity

Table 1

Partial Derivatives of Functions of Two Variables (3 of 14)

If we concentrate on the highlighted column of the table, which corresponds to a relative humidity of H = 70%, we are considering the heat index as a function of the single variable T for a fixed value of H. Let's write $g(T) = f(T, 70)$.

Then $g(T)$ describes how the heat index I increases as the actual temperature T increases when the relative humidity is 70%. The derivative of g when T = 96°F is the rate of change of I with respect to T when T = 96°F:

$$g'(96) = \lim_{h \to 0} \frac{g(96 + h) - g(96)}{h} = \lim_{h \to 0} \frac{f(96 + h,\, 70) - f(96,\, 70)}{h}$$

Partial Derivatives of Functions of Two Variables (4 of 14)

We can approximate $g'(96)$ using the values in Table 1 by taking h = 2 and −2:

$$g'(96) \approx \frac{g(98) - g(96)}{2} = \frac{f(98,70) - f(96,70)}{2} = \frac{133 - 125}{2} = 4$$

$$g'(96) \approx \frac{g(94) - g(96)}{-2} = \frac{f(94,70) - f(96,70)}{-2} = \frac{118 - 125}{-2} = 3.5$$

Averaging these values, we can say that the derivative $g'(96)$ is approximately 3.75.

Partial Derivatives of Functions of Two Variables (5 of 14)

This means that, when the actual temperature is 96°F and the relative humidity is 70%, the apparent temperature (heat index) rises by about 3.75°F for every degree that the actual temperature rises.

Now let's look at the highlighted row in Table 1, which corresponds to a fixed temperature of T = 96°F.

Relative humidity (%)

Actual temperature (°F)

T \ H	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	125	130	135	141	146
98	114	118	123	127	133	138	144	150	157
100	119	124	129	135	141	147	154	161	168

Heat index *I* as a function of temperature and humidity

Table 1

Partial Derivatives of Functions of Two Variables (7 of 14)

The numbers in this row are values of the function $G(H) = f(96, H)$, which describes how the heat index increases as the relative humidity H increases when the actual temperature is $T = 96°$F.

The derivative of this function when $H = 70\%$ is the rate of change of I with respect to H when $H = 70\%$:

$$G'(70) = \lim_{h \to 0} \frac{G(70 + h) - G(70)}{h} = \lim_{h \to 0} \frac{f(96, 70 + h) - f(96, 70)}{h}$$

Partial Derivatives of Functions of Two Variables (8 of 14)

By taking h = 5 and −5, we approximate $G'(70)$ using the tabular values:

$$G'(70) \approx \frac{G(75) - G(70)}{5} = \frac{f(96,75) - f(96,70)}{5} = \frac{130 - 125}{5} = 1$$

$$G'(70) \approx \frac{G(65) - G(70)}{-5} = \frac{f(96,65) - f(96,70)}{-5} = \frac{121 - 125}{-5} = 0.8$$

By averaging these values we get the estimate $G'(70) \approx 0.9$. This says that, when the temperature is 96°F and the relative humidity is 70%, the heat index rises about 0.9°F for every percent that the relative humidity rises.

Partial Derivatives of Functions of Two Variables (9 of 14)

In general, if f is a function of two variables x and y, suppose we let only x vary while keeping y fixed, say $y = b$, where b is a constant.

Then we are really considering a function of a single variable x, namely, $g(x) = f(x, b)$. If g has a derivative at a, then we call it the **partial derivative of *f* with respect to *x* at (*a*, *b*)** and denote it by $f_x(a, b)$. Thus

偏导数

1 $$f_x(a,b) = g'(a) \quad \text{where} \quad g(x) = f(x,b)$$

Partial Derivatives of Functions of Two Variables (10 of 14)

By the definition of a derivative, we have

$$g'(a) = \lim_{h \to 0} \frac{g(a+h) - g(a)}{h}$$

and so Equation 1 becomes

2 $$f_x(a,b) = \lim_{h \to 0} \frac{f(a+h,\, b) - f(a,\, b)}{h}$$

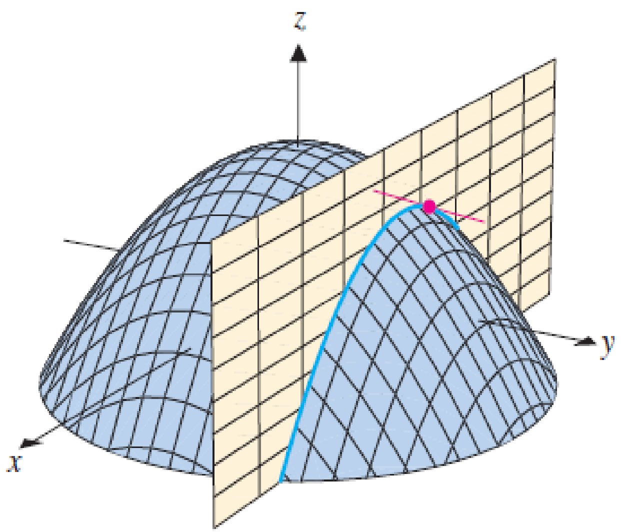


FIGURE 12.20a
Intersection of the surface
$z = f(x, y)$ with the plane $y = b$

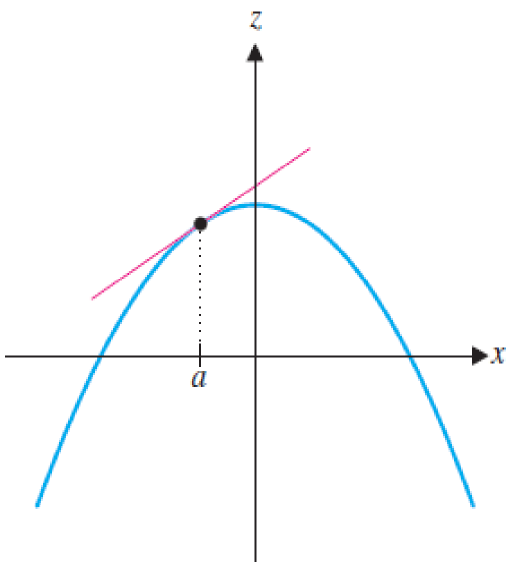


FIGURE 12.20b
The curve $z = f(x, b)$

Partial Derivatives of Functions of Two Variables (11 of 14)

Similarly, the **partial derivative of *f* with respect to *y* at (*a*, *b*)**, denoted by $f_y(a, b)$, is obtained by keeping x fixed ($x = a$) and finding the ordinary derivative at b of the function $G(y) = f(a, y)$:

$$f_y(a,b) = \lim_{h \to 0} \frac{f(a, b+h) - f(a,b)}{h} \tag{3}$$

With this notation for partial derivatives, we can write the rates of change of the heat index I with respect to the actual temperature T and relative humidity H when $T = 96$°F and $H = 70$% as follows:

$$f_T(96, 70) \approx 3.75 \qquad f_H(96, 70) \approx 0.9$$

Partial Derivatives of Functions of Two Variables (12 of 14)

If we now let the point (a, b) vary in Equations 2 and 3, f_x and f_y become functions of two variables.

4 Definition If f is a function of two variables, its **partial derivatives** are the functions f_x and f_y defined by

$$f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Partial Derivatives of Functions of Two Variables (13 of 14)

There are many alternative notations for partial derivatives. For instance, instead of f_x we can write f_1 or D_1f (to indicate differentiation with respect to the *first* variable) or $\partial f / \partial x$.

But here $\partial f/\partial x$ can't be interpreted as a ratio of differentials.

Notations for Partial Derivatives If $z = f(x, y)$, we write

$$f_x(x,y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x,y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

$$f_y(x,y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x,y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

Partial Derivatives of Functions of Two Variables (14 of 14)

To compute partial derivatives, all we have to do is remember from Equation 1 that the partial derivative with respect to x is just the *ordinary* derivative of the function g of a single variable that we get by keeping y fixed.

Thus we have the following rule.

Rule for Finding Partial Derivatives of $z = f(x, y)$

1. To find $f_{x,}$ regard y as a constant and differentiate $f(x, y)$ with respect to x.

2. To find $f_{y,}$ regard x as a constant and differentiate $f(x, y)$ with respect to y.

Example 1

If $f(x,y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ and $f_y(2, 1)$.

Solution:

Holding y constant and differentiating with respect to x, we get

$$f_x(x, y) = 3x^2 + 2xy^3$$

and so $$f_x(2, 1) = 3 \cdot 2^2 + 2 \cdot 2 \cdot 1^3 = 16$$

Holding x constant and differentiating with respect to y, we get

$$f_y(x, y) = 3x^2y^2 - 4y$$

$$f_y(2, 1) = 3 \cdot 2^2 \cdot 1^2 - 4 \cdot 1 = 8$$

Interpretations of Partial Derivatives

Interpretations of Partial Derivatives (1 of 4)

To give a geometric interpretation of partial derivatives, we know that the equation $z = f(x, y)$ represents a surface S (the graph of f). If $f(a, b) = c$, then the point $P(a, b, c)$ lies on S.

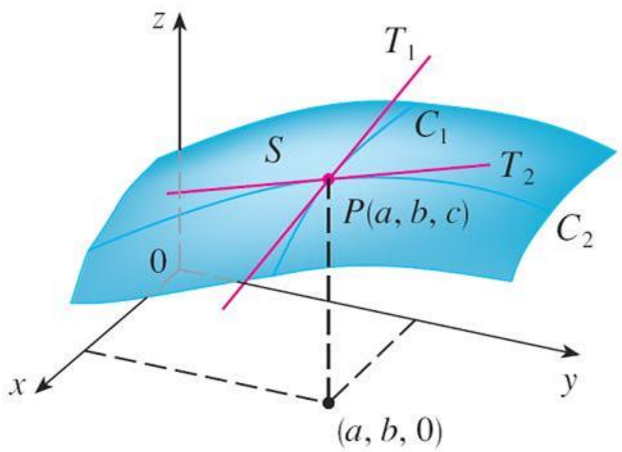
By fixing $y = b$, we are restricting our attention to the curve C_1 in which the vertical plane $y = b$ intersects S. (In other words, C_1 is the trace of S in the plane $y = b$.)

Interpretations of Partial Derivatives (2 of 4)

Likewise, the vertical plane $x = a$ intersects S in a curve C_2. Both of the curves C_1 and C_2 pass through the point P. (See Figure 1.)

Note that the curve C_1 is the graph of the function $g(x) = f(x, b)$, so the slope of its tangent T_1 at P is $g'(a) = f_x(a, b)$.

The curve C_2 is the graph of the function $G(y) = f(a, y)$, so the slope of its tangent T_2 at P is $G'(b) = f_y(a, b)$.



The partial derivatives of f at (a, b) are the slopes of the tangents to C_1 and C_2.

Figure 1

Interpretations of Partial Derivatives (3 of 4)

Thus the partial derivatives $f_x(a, b)$ and $f_y(a, b)$ can be interpreted geometrically as the slopes of the tangent lines at $P(a, b, c)$ to the traces C_1 and C_2 of S in the planes $y = b$ and $x = a$.

As we have seen in the case of the heat index function at the beginning of this section, partial derivatives can also be interpreted as *rates of change*.

If $z = f(x, y)$, then $\partial z/\partial x$ represents the rate of change of z with respect to x when y is fixed. Similarly, $\partial z/\partial y$ represents the rate of change of z with respect to y when x is fixed.

Example 3

If $f(x,y) = 4 - x^2 - 2y^2$, find $f_x(1, 1)$ and $f_y(1, 1)$ and interpret these numbers as slopes.

Solution:

We have

$$f_x(x, y) = -2x \qquad f_y(x, y) = -4y$$

$$f_x(1, 1) = -2 \qquad f_y(1, 1) = -4$$

Example 3 – Solution (1 of 2)

The graph of f is the paraboloid $z = 4 - x^2 - 2y^2$ and the vertical plane $y = 1$ intersects it in the parabola $z = 2 - x^2$, $y = 1$. (As in the preceding discussion, we label it C_1 in Figure 2.)

The slope of the tangent line to this parabola at the point (1, 1, 1) is $f_x(1, 1) = -2$.

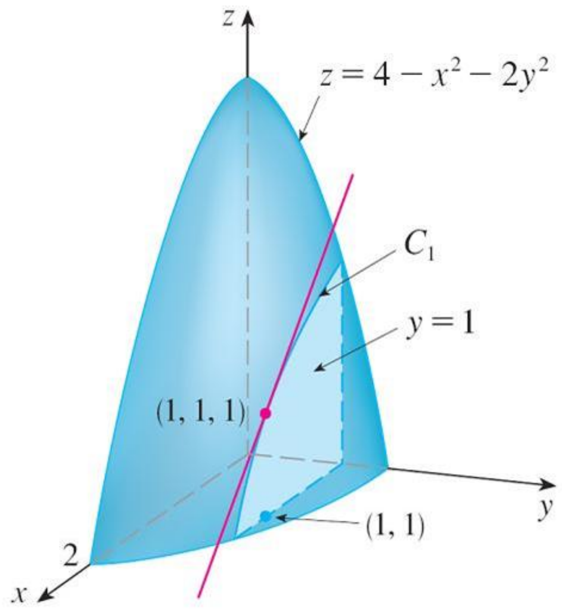


Figure 2

Example 3 – Solution (2 of 2)

Similarly, the curve C_2 in which the plane $x = 1$ intersects the paraboloid is the parabola $z = 3 - 2y^2$, $x = 1$, and the slope of the tangent line at (1, 1, 1) is $f_y(1, 1) = -4$. (See Figure 3.)

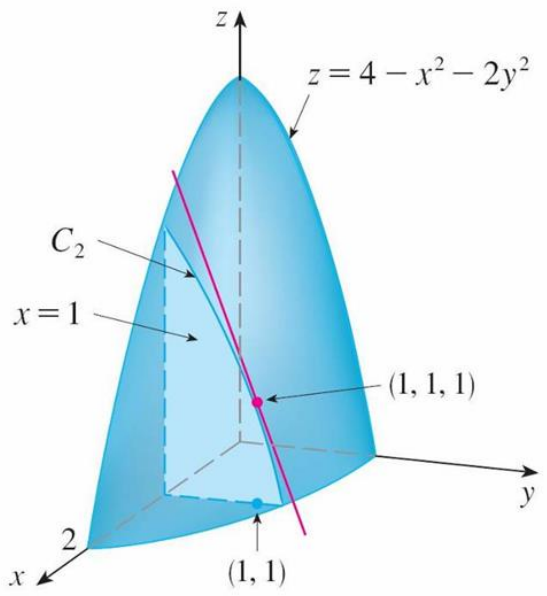


Figure 3

Interpretations of Partial Derivatives (4 of 4)

As we have seen in the case of the heat index function at the beginning of this section, partial derivatives can also be interpreted as *rates of change*.

If $z = f(x, y)$, then $\partial z/\partial x$ represents the rate of change of z with respect to x when y is fixed. Similarly, $\partial z/\partial y$ represents the rate of change of z with respect to y when x is fixed.

Functions of Three or More Variables

Functions of Three or More Variables (1 of 2)

Partial derivatives can also be defined for functions of three or more variables. For example, if *f* is a function of three variables *x*, *y*, and *z*, then its partial derivative with respect to *x* is defined as

$$f_x(x, y, z) = \lim_{h \to 0} \frac{f(x+h, y, z) - f(x, y, z)}{h}$$

and it is found by regarding *y* and *z* as constants and differentiating *f*(*x*, *y*, *z*) with respect to *x*.

Functions of Three or More Variables (2 of 2)

If $w = f(x, y, z)$, then $f_x = \partial w/\partial x$ can be interpreted as the rate of change of w with respect to x when y and z are held fixed. But we can't interpret it geometrically because the graph of f lies in four-dimensional space.

In general, if u is a function of n variables, $u = f(x_1, x_2, \ldots, x_n)$, its partial derivative with respect to the i th variable x_i is

$$\frac{\partial u}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{h}$$

and we also write

$$\frac{\partial u}{\partial x_i} = \frac{\partial f}{\partial x_i} = f_{x_i} = f_i = D_i f$$

Example 6

Find f_x, f_y, and f_z if $f(x, y, z) = e^{xy} \ln z$.

Solution:

Holding y and z constant and differentiating with respect to x, we have

$$f_x = ye^{xy} \ln z$$

Similarly,

$$f_y = xe^{xy} \ln z \qquad \text{and} \qquad f_z = \frac{e^{xy}}{z}$$

Higher Derivatives

Higher Derivatives (1 of 4)

If f is a function of two variables, then its partial derivatives f_x and f_y are also functions of two variables, so we can consider their partial derivatives $(f_x)_x$, $(f_x)_y$, $(f_y)_x$, and $(f_y)_y$, which are called the **second partial derivatives** of f.

If $z = f(x, y)$, we use the following notation:

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

Higher Derivatives (2 of 4)

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

Thus the notation f_{xy} $\left(\text{or } \partial^2 f / \partial y \partial x\right)$ means that we first differentiate with respect to x and then with respect to y, whereas in computing f_{yx} the order is reversed.

Example 7

Find the second partial derivatives of

$$f(x, y) = x^3 + x^2y^3 - 2y^2$$

Solution:

In Example 1 we found that

$$f_x(x, y) = 3x^2 + 2xy^3 \qquad f_y(x, y) = 3x^2y^2 - 4y$$

Therefore

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x}(3x^2 + 2xy^3) \\ &= 6x + 2y^3 \end{aligned}$$

Example 7 – Solution

$$f_{xy} = \frac{\partial}{\partial y}(3x^2 + 2xy^3)$$

$$= 6xy^2$$

$$f_{yx} = \frac{\partial}{\partial x}(3x^2y^2 - 4y)$$

$$= 6xy^2$$

$$f_{yy} = \frac{\partial}{\partial y}(3x^2y^2 - 4y)$$

$$= 6x^2y - 4$$

Higher Derivatives (3 of 4)

Notice that $f_{xy} = f_{yx}$ in Example 7. This is not just a coincidence. It turns out that the mixed partial derivatives f_{xy} and f_{yx} are equal for most functions that one meets in practice.

The following theorem, which was discovered by the French mathematician Alexis Clairaut (1713−1765), gives conditions under which we can assert that $f_{xy} = f_{yx}$.

Clairaut's Theorem Suppose f is defined on a disk D that contains the point (a, b). If the functions f_{xy} and f_{yx} are both continuous on D, then

$$f_{xy}(a,b) = f_{yx}(a,b)$$

Higher Derivatives (4 of 4)

Partial derivatives of order 3 or higher can also be defined. For instance,

$$f_{xyy} = (f_{xy})_y = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial y \partial x}\right) = \frac{\partial^3 f}{\partial y^2 \partial x}$$

and using Clairaut's Theorem it can be shown that $f_{xyy} = f_{yxy} = f_{yyx}$ if these functions are continuous.

✓Gradient (梯度)

We denote the gradient of a function f by **grad** f or ∇f (read "del f").

DEFINITION 6.2

The **gradient** of $f(x, y)$ is the vector-valued function

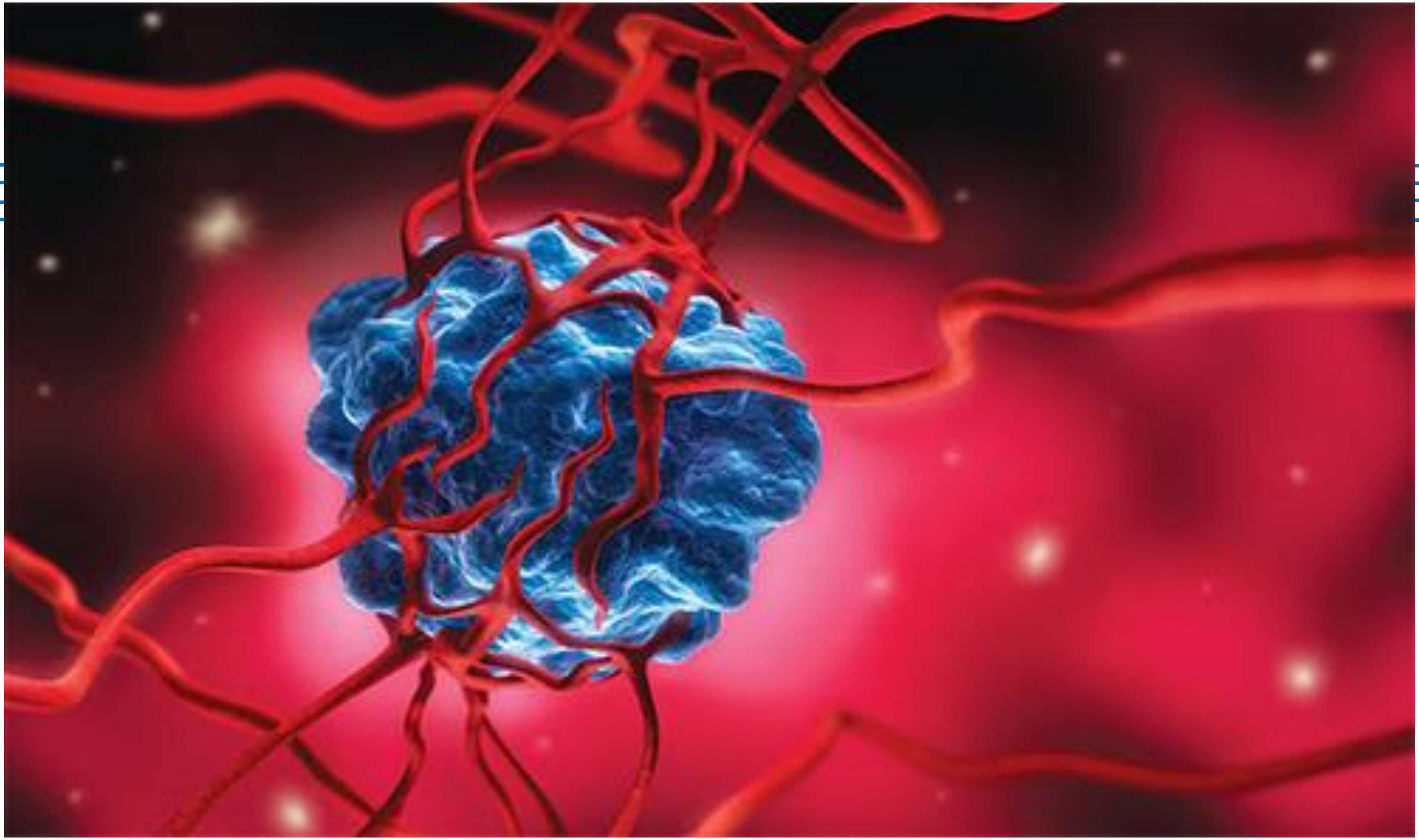
$$\nabla f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j},$$

provided both partial derivatives exist.

$$f(\mathbf{p}+\mathbf{h}) = f(\mathbf{p}) + \nabla f(\mathbf{p})\bullet\mathbf{h} + \varepsilon(\mathbf{h})\bullet\mathbf{h}$$

Where $\varepsilon(\mathbf{h}) \to 0$ as $\mathbf{h} \to 0$.

15 Multiple Integrals



15.1 Double Integrals over Rectangles

Review of the Definite Integral

Review of the Definite Integral (1 of 2)

First let's recall the basic facts concerning definite integrals of functions of a single variable.

If $f(x)$ is defined for $a \le x \le b$, we start by dividing the interval $[a, b]$ into n subintervals $[x_{i-1}, x_i]$ of equal width $\Delta x = (b-a)/n$ and we choose sample points x_i^* in these subintervals. Then we form the Riemann sum

$$\textbf{1} \qquad \sum_{i=1}^{n} f\left(x_i^*\right)\Delta x$$

and take the limit of such sums as $n \to \infty$ to obtain the definite integral of f from a to b:

$$\textbf{2} \qquad \int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right)\Delta x$$

Review of the Definite Integral (2 of 2)

In the special case where $f(x) \geq 0$, the Riemann sum can be interpreted as the sum of the areas of the approximating rectangles in Figure 1, and $\int_a^b f(x)\,dx$ represents the area under the curve $y = f(x)$ from a to b.

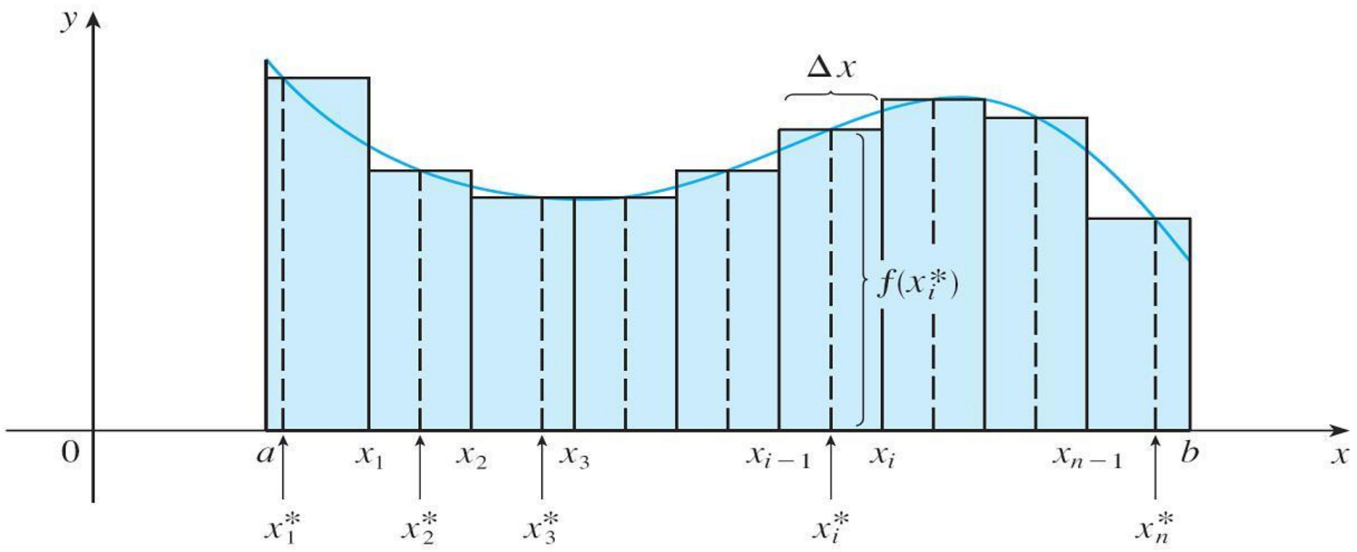


Figure 1

Volumes and Double Integrals

In a similar manner we consider a function *f* of two variables defined on a closed rectangle

$$R = [a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b,\ c \leq y \leq d\}$$

and we first suppose that $f(x, y) \geq 0$. The graph of *f* is a surface with equation $z = f(x, y)$. Let *S* be the solid that lies above *R* and under the graph of *f*, that is,

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq z \leq f(x, y),\ (x, y) \in R\}$$

(See Figure 2.)

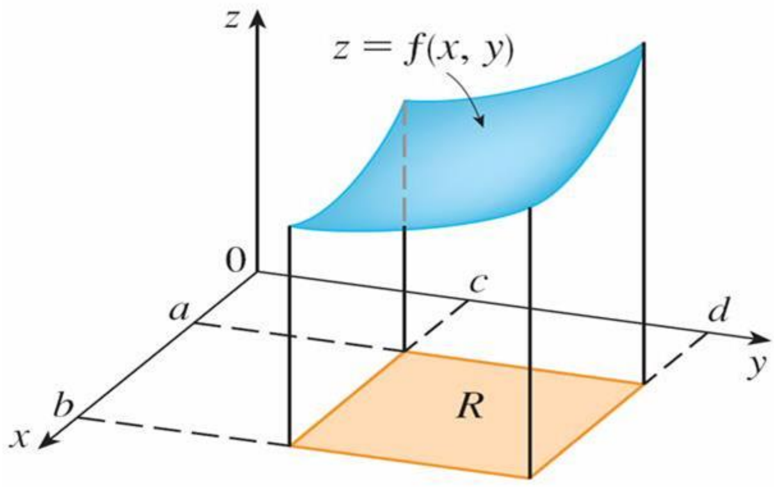


Figure 2

Volumes and Double Integrals (2 of 13)

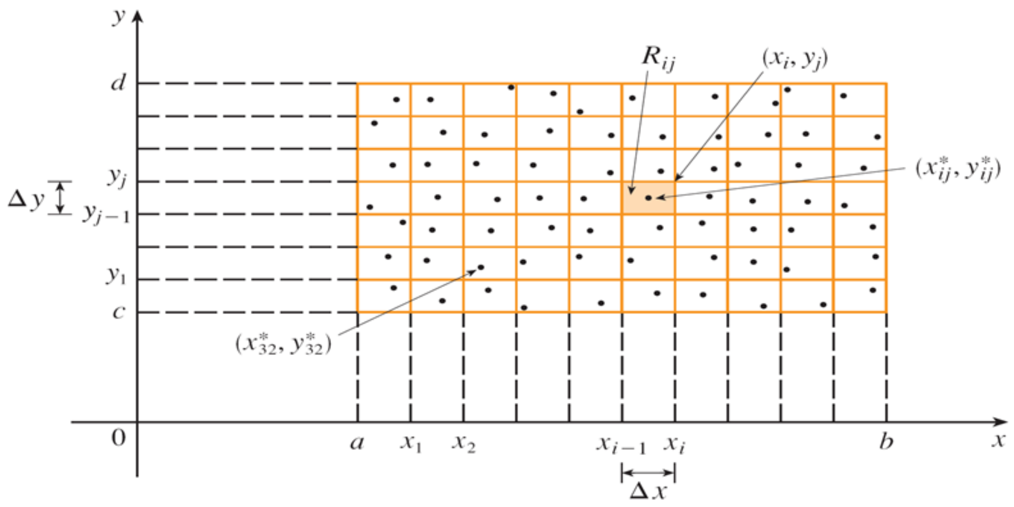
Our goal is to find the volume of *S*.

The first step is to divide the rectangle *R* into subrectangles. We accomplish this by dividing the interval $[a, b]$ into m subintervals $[x_{i-1}, x_i]$ of equal width $\Delta x = (b - a)/m$ and dividing $[c, d]$ into n subintervals $[y_{j-1}, y_j]$ of equal width $\Delta y = (d - c)/n$.

By drawing lines parallel to the coordinate axes through the endpoints of these subintervals, as in Figure 3, we form the subrectangles

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] = \{(x, y) \mid x_{i-1} \le x \le x_i,\ y_{j-1} \le y \le y_j\}$$

each with area $\Delta A = \Delta x\, \Delta y$.



Dividing *R* into subrectangles

Figure 3

If we choose a **sample point** (x_{ij}^*, y_{ij}^*) in each R_{ij}, then we can approximate the part of S that lies above each R_{ij} by a thin rectangular box (or "column") with base R_{ij} and height $f\left(x_{ij}^*, y_{ij}^*\right)$ 燼s shown in Figure 4.

The volume of this box is the height of the box times the area of the base rectangle:

$$f(x_{ij}^*, y_{ij}^*)\Delta A$$

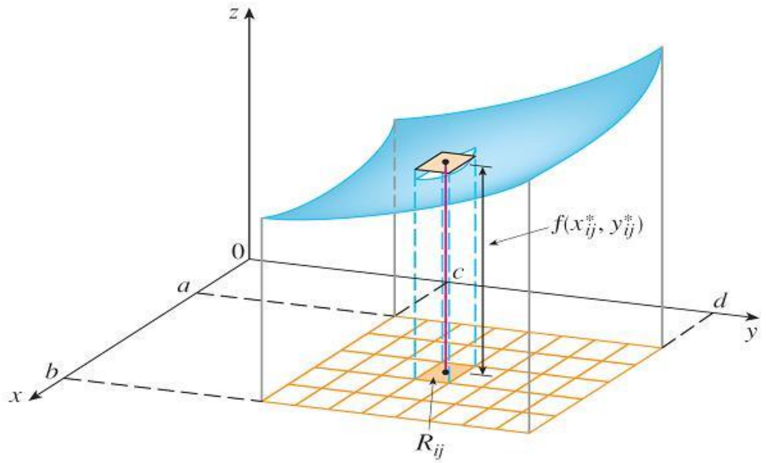


Figure 4

Volumes and Double Integrals (5 of 13)

If we follow this procedure for all the rectangles and add the volumes of the corresponding boxes, we get an approximation to the total volume of *S*:

3 $$V \approx \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \Delta A$$

(See Figure 5.) This double sum means that for each subrectangle we evaluate *f* at the chosen point and multiply by the area of the subrectangle, and then we add the results.

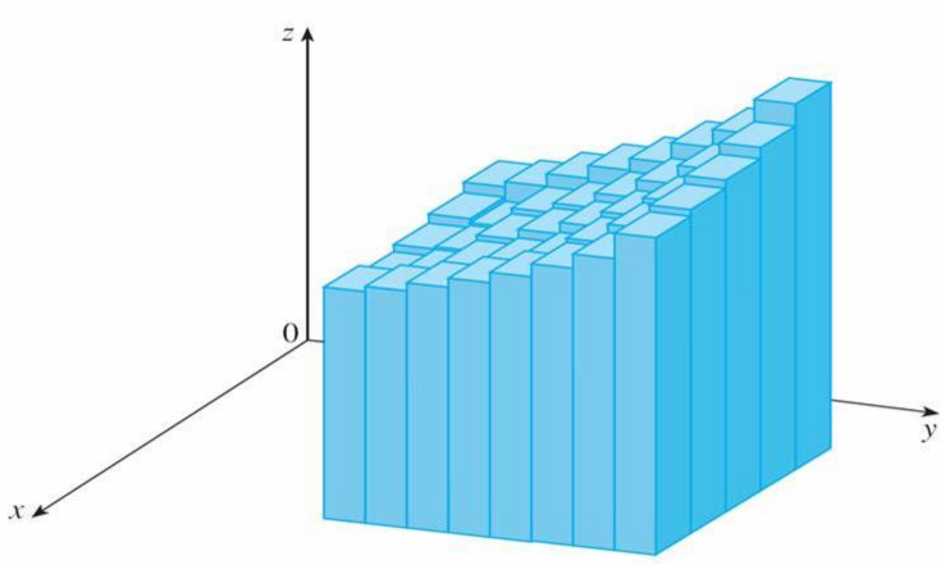


Figure 5

Volumes and Double Integrals (6 of 13)

Our intuition tells us that the approximation given in (3) becomes better as *m* and *n* become larger and so we would expect that

$$4 \quad V = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^{*}, y_{ij}^{*})\Delta A$$

We use the expression in Equation 4 to define the **volume** of the solid *S* that lies under the graph of *f* and above the rectangle *R*.

Limits of the type that appear in Equation 4 occur frequently, not just in finding volumes but in a variety of other situations even when *f* is not a positive function. So we make the following definition.

5 Definition The **double integral** of *f* over the rectangle *R* is

$$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\,\Delta A$$

if this limit exists.

Volumes and Double Integrals (8 of 13)

The precise meaning of the limit in Definition 5 is that for every number $\varepsilon > 0$ there is an integer N such that

$$\left| \iint_R f(x, y)\, dA - \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \Delta A \right| < \varepsilon$$

for all integers m and n greater than N and for any choice of sample points $\left(x_{ij}^*, y_{ij}^*\right)$ in R_{ij}.

A function f is called **integrable** if the limit in Definition 5 exists.

Volumes and Double Integrals (9 of 13)

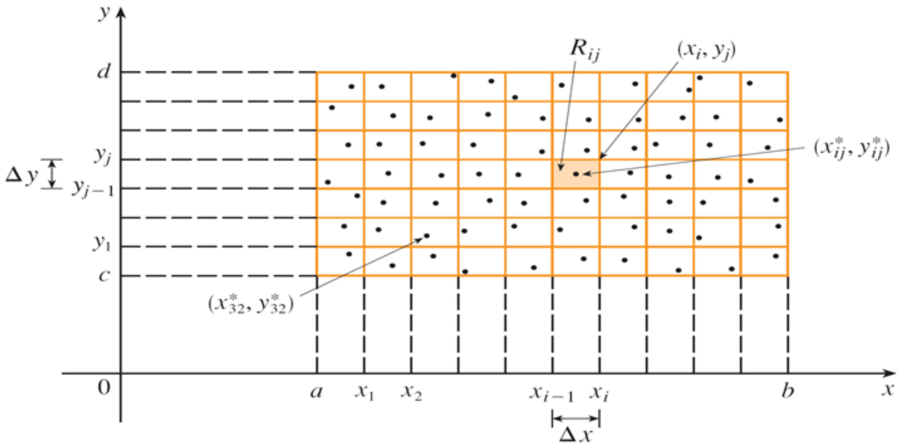
It is shown in courses on advanced calculus that all continuous functions are integrable. In fact, the double integral of *f* exists provided that *f* is "not too discontinuous."

In particular, if *f* is bounded on *R*, [that is, there is a constant *M* such that $|f(x, y)| \leq M$ for all (x, y) in *R*], and *f* is continuous there, except on possibly a finite number of smooth curves, then *f* is integrable over *R*.

Volumes and Double Integrals (10 of 13)

The sample point (x_{ij}^*, y_{ij}^*) can be chosen to be any point in the subrectangle R_{ij}, but if we choose it to be the upper right-hand corner of R_{ij} [namely (x_i, y_j), see Figure 3], then the expression for the double integral looks simpler:

6 $$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i, y_j)\,\Delta A$$



Dividing R into subrectangles

Figure 3

Volumes and Double Integrals (11 of 13)

By comparing Definitions 4 and 5, we see that a volume can be written as a double integral:

If $f(x, y) \geq 0$, then the volume V of the solid that lies above the rectangle R and below the surface $z = f(x, y)$ is

$$V = \iint_R f(x, y)\, dA$$

Volumes and Double Integrals (12 of 13)

The sum in Definition 5,

$$\sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^{*}, y_{ij}^{*})\Delta A$$

is called a **double Riemann sum** and is used as an approximation to the value of the double integral. [Notice how similar it is to the Riemann sum in (1) for a function of a single variable.]

Volumes and Double Integrals (13 of 13)

If *f* happens to be a *positive* function, then the double Riemann sum represents the sum of volumes of columns, as in Figure 5, and is an approximation to the volume under the graph of *f*.

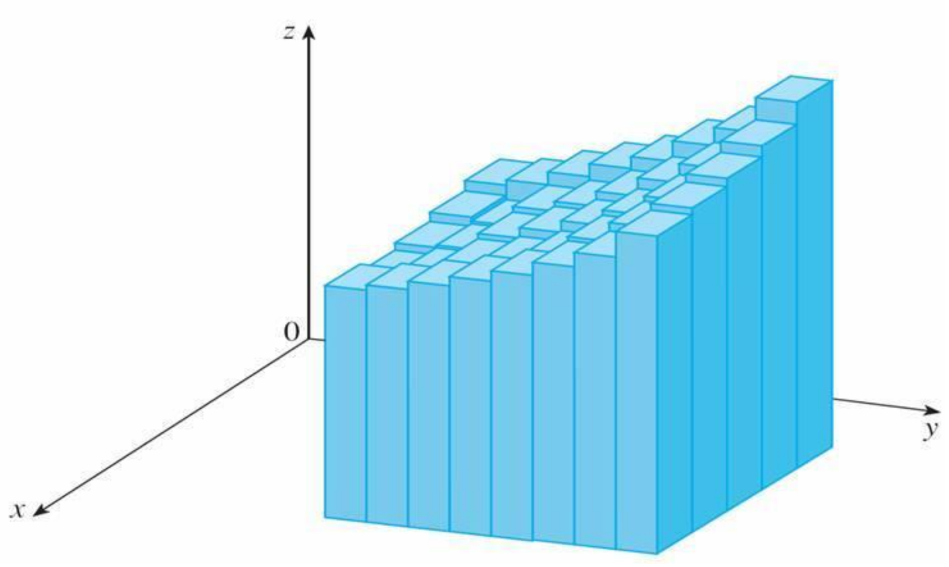


Figure 5

Example 1

Estimate the volume of the solid that lies above the square $R = [0, 2] \times [0, 2]$ and below the elliptic paraboloid $z = 16 - x^2 - 2y^2$. Divide R into four equal squares and choose the sample point to be the upper right corner of each square R_{ij}. Sketch the solid and the approximating rectangular boxes.

Example 1 – Solution (1 of 3)

The squares are shown in Figure 6.

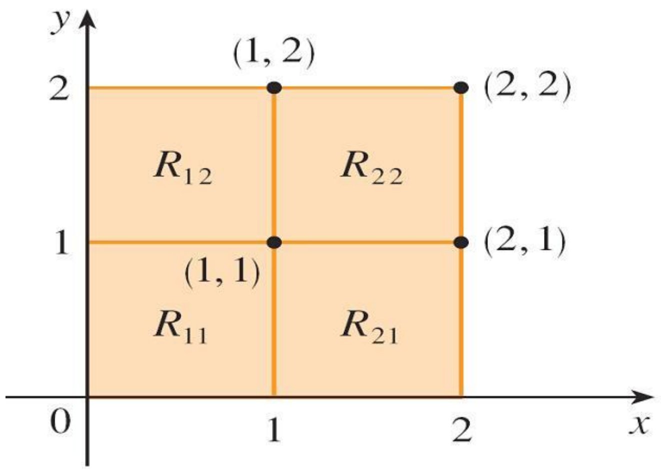


Figure 6

The paraboloid is the graph of $f(x, y) = 16 - x^2 - 2y^2$ and the area of each square is $\Delta A = 1$.

Example 1 – Solution (2 of 3)

Approximating the volume by the Riemann sum with $m = n = 2$, we have

$$
\begin{aligned}
V &\approx \sum_{i=1}^{2}\sum_{j=1}^{2} f(x_i, y_j)\Delta A \\
&= f(1,1)\Delta A + f(1,2)\Delta A + f(2,1)\Delta A + f(2,2)\Delta A \\
&= 13(1) + 7(1) + 10(1) + 4(1) \\
&= 34
\end{aligned}
$$

Example 1 – Solution (3 of 3)

This is the volume of the approximating rectangular boxes shown in Figure 7.

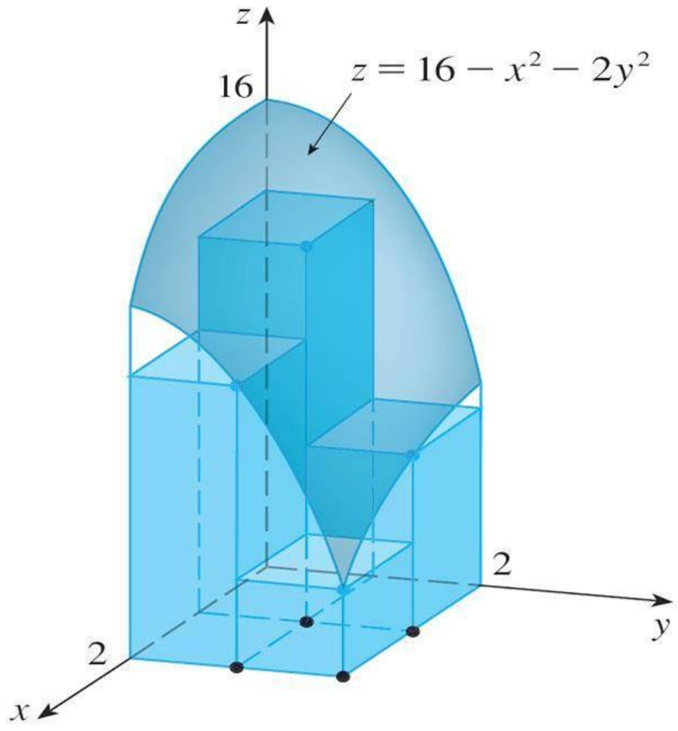


Figure 7

Iterated Integrals

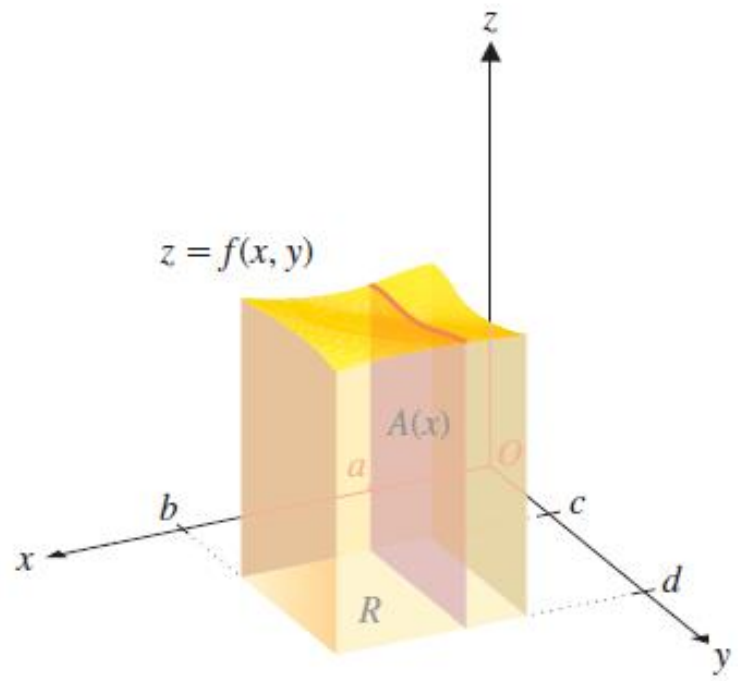


FIGURE 13.6a
Slicing the solid parallel to the *yz*-plane

$$V = \int_a^b A(x)\,dx.$$

$$A(x) = \int_c^d f(x, y)\,dy.$$

$$V = \int_a^b A(x)\,dx = \int_a^b \left[\int_c^d f(x, y)\,dy\right] dx.$$

Iterated Integrals (1 of 7)

Suppose that f is a function of two variables that is integrable on the rectangle $R = [a, b] \times [c, d]$.

We use the notation $\int_c^d f(x, y)\, dy$ to mean that x is held fixed and $f(x, y)$ is integrated with respect to y from $y = c$ to $y = d$. This procedure is called *partial integration with respect to y*. (Notice its similarity to partial differentiation.)

Now $\int_c^d f(x, y)\, dy$ is a number that depends on the value of x, so it defines a function of x:

$$A(x) = \int_c^d f(x,y)\, dy$$

Iterated Integrals (2 of 7)

If we now integrate the function A with respect to x from $x = a$ to $x = b$, we get

$$\mathbf{7} \qquad \int_a^b A(x)\,dx = \int_a^b \left[\int_c^d f(x, y)\,dy \right] dx$$

The integral on the right side of Equation 7 is called an **iterated integral**. Usually the brackets are omitted. Thus

$$\mathbf{8} \qquad \int_a^b \int_c^d f(x,y)\,dy\ dx = \int_a^b \left[\int_c^d f(x, y)\,dy \right] dx$$

means that we first integrate with respect to y (holding x fixed) from $y = c$ to $y = d$, and then we integrate the resulting function of x with respect to x from $x = a$ to $x = b$.

Iterated Integrals (3 of 7)

Similarly, the iterated integral

9 $$\int_c^d \int_a^b f(x, y)\, dy\, dx = \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy$$

means that we first integrate with respect to x (holding y fixed) from $x = a$ to $x = b$ and then we integrate the resulting function of y with respect to y from $y = c$ to $y = d$.

Notice that in both Equations 8 and 9 we work *from the inside out*.

Example 4

Evaluate the iterated integrals.

(a) $\int_0^3 \int_1^2 x^2 y \; dy \; dx$ (b) $\int_1^2 \int_0^3 x^2 y \; dx \; dy$

Solution:

(a) Regarding x as a constant, we obtain

$$\int_1^2 x^2 y \; dy = \left[x^2 \frac{y^2}{2} \right]_{y=1}^{y=2}$$

$$= x^2 \left(\frac{2^2}{2} \right) - x^2 \left(\frac{1^2}{2} \right)$$

$$= \tfrac{3}{2} x^2$$

Example 4 – Solution (1 of 2)

Thus the function A in the preceding discussion is given by $A(x) = \frac{3}{2}x^2$ in this example.

We now integrate this function of x from 0 to 3:

$$\begin{aligned}
\int_0^3 \int_1^2 x^2 y \, dy \, dx &= \int_0^3 \left[\int_1^2 x^2 y \, dy \right] dx \\
&= \int_0^3 \tfrac{3}{2} x^2 \, dx \\
&= \left. \frac{x^3}{2} \right]_0^3 \\
&= \frac{27}{2}
\end{aligned}$$

Example 4 – Solution (2 of 2)

(b) Here we first integrate with respect to x, regarding y as a constant:

$$\int_1^2 \int_0^3 x^2 y \, dx \, dy = \int_1^2 \left[\int_0^3 x^2 y \, dx \right] dy$$

$$= \int_1^2 \left[\frac{x^3}{3} y \right]_{x=0}^{x=3} dy$$

$$= \int_1^2 9y \, dy$$

$$= 9 \frac{y^2}{2} \Big]_1^2 = \frac{27}{2}$$

Iterated Integrals (4 of 7)

Notice that in Example 4 we obtained the same answer whether we integrated with respect to y or x first.

In general, it turns out (see Theorem 10) that the two iterated integrals in Equations 8 and 9 are always equal; that is, the order of integration does not matter. (This is similar to Clairaut's Theorem on the equality of the mixed partial derivatives.)

Iterated Integrals (5 of 7)

The following theorem gives a practical method for evaluating a double integral by expressing it as an iterated integral (in either order).

10 Fubini's Theorem If *f* is continuous on the rectangle

$$R = \{(x, y) \mid a \le x \le b,\ c \le y \le d\},$$

then

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy$$

More generally, this is true if we assume that *f* is bounded on *R*, f is discontinuous only on a finite number of smooth curves, and the iterated integrals exist.

Iterated Integrals (6 of 7)

In the special case where $f(x, y)$ can be factored as the product of a function of x only and a function of y only, the double integral of f can be written in a particularly simple form.

To be specific, suppose that $f(x, y) = g(x)h(y)$ and $R = [a, b] \times [c, d]$. Then Fubini's Theorem gives

$$\iint_R f(x, y)\, dA = \int_c^d \int_a^b g(x)h(y)\, dx\, dy = \int_c^d \left[\int_a^b g(x)h(y)\, dx \right] dy$$

Iterated Integrals (7 of 7)

In the inner integral, *y* is a constant, so *h*(*y*) is a constant and we can write

$$\int_c^d \left[\int_a^b g(x)h(y)\,dx\right] dy = \int_c^d \left[h(y)\left(\int_a^b g(x)\,dx\right)\right] dy = \int_a^b g(x)\,dx \int_c^d h(y)\,dy$$

since $\int_a^b g(x)\,dx$ is a constant.

Therefore, in this case the double integral of *f* can be written as the product of two single integrals:

11 $$\iint_R g(x)h(y)\,dA = \int_a^b g(x)\,dx \int_c^d h(y)\,dy \qquad \text{where } R = [a,b]\times[c,d]$$

15.2 Double Integrals over General Regions

Double Integrals over General Regions

For single integrals, the region over which we integrate is always an interval.

But for double integrals, we want to be able to integrate a function not just over rectangles but also over regions of more general shape.

General Regions

General Regions (1 of 12)

Consider a general region *D* like the one illustrated in Figure 1.

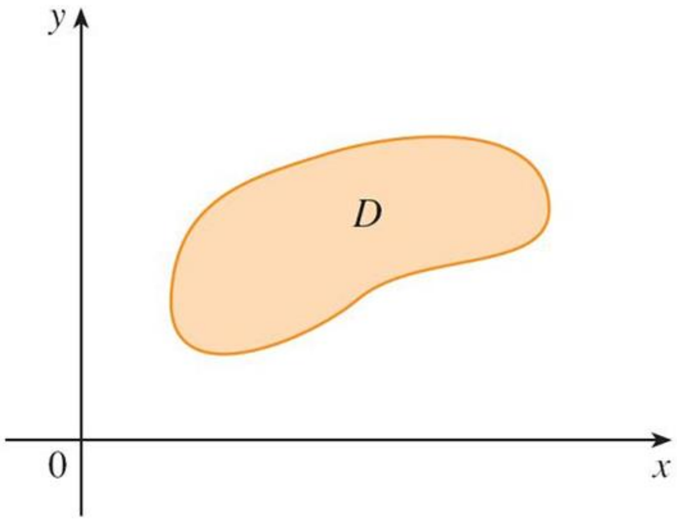


Figure 1

General Regions (2 of 12)

We suppose that *D* is a bounded region, which means that *D* can be enclosed in a rectangular region *R* as in Figure 2.

In order to integrate a function *f* over *D* we define a new function *F* with domain *R* by

1 $$F(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \text{ is in } D \\ 0 & \text{if } (x, y) \text{ is in } R \text{ but not in } D \end{cases}$$

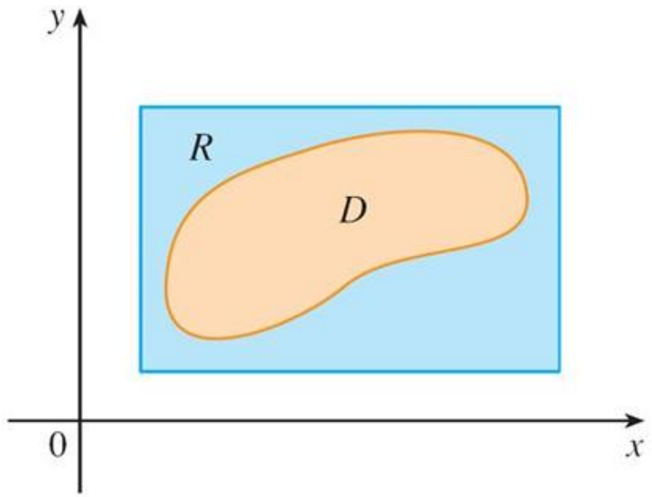


Figure 2

General Regions (3 of 12)

If F is integrable over R, then we define the **double integral of *f* over *D*** by

2 $$\iint_D f(x, y)\, dA = \iint_R F(x, y)\, dA \qquad \text{where } F \text{ is given by Equation 1}$$

Definition 2 makes sense because R is a rectangle and so $\iint_R F(x, y)\, dA$ has been previously defined.

General Regions (4 of 12)

The procedure that we have used is reasonable because the values of $F(x, y)$ are 0 when (x, y) lies outside D and so they contribute nothing to the integral. This means that it doesn't matter what rectangle R we use as long as it contains D.

In the case where $f(x, y) \geq 0$, we can still interpret $\iint_D f(x, y)\, dA$ as the volume of the solid that lies above D and under the surface $z = f(x, y)$ (the graph of f).

You can see that this is reasonable by comparing the graphs of f and F in Figures 3 and 4 and remembering that $\iint_R F(x, y)\, dA$ is the volume under the graph of F.

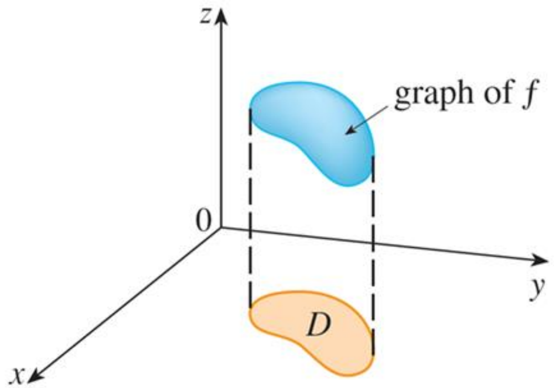


Figure 3

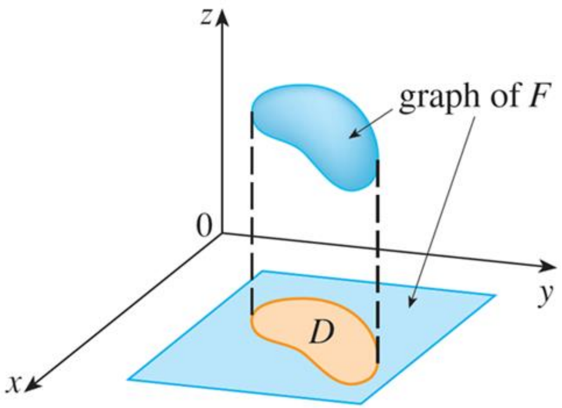


Figure 4

General Regions (6 of 12)

Figure 4 also shows that *F* is likely to have discontinuities at the boundary points of *D*.

Nonetheless, if *f* is continuous on *D* and the boundary curve of *D* is "well behaved", then it can be shown that $\iint_R F(x, y)\, dA$ exists and therefore $\iint_D f(x, y)\, dA$ exists.

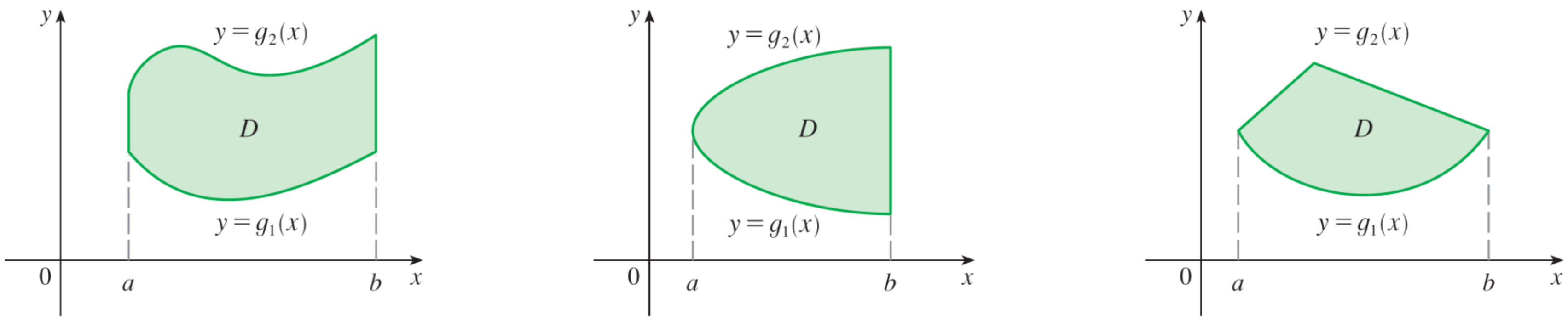
In particular, this is the case for **type I** and **type II** regions.

General Regions (7 of 12)

A plane region D is said to be of **type I** if it lies between the graphs of two continuous functions of x, that is,

$$D = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$$

where g_1 and g_2 are continuous on [a, b]. Some examples of type I regions are shown in Figure 5.



Some type I regions

Figure 5

General Regions (8 of 12)

In order to evaluate $\iint_D f(x, y)\, dA$ when D is a region of type I, we choose a rectangle $R = [a, b] \times [c, d]$ that contains D, as in Figure 6, and we let F be the function given by Equation 1; that is, F agrees with f on D and F is 0 outside D.

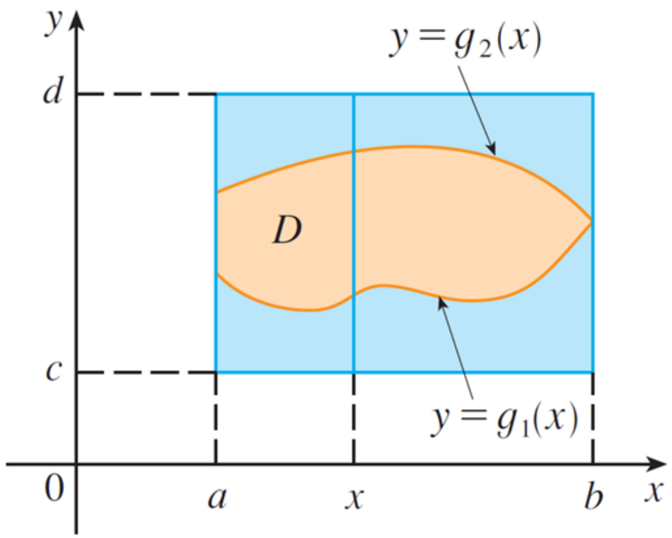


Figure 6

General Regions (9 of 12)

Then, by Fubini's Theorem,

$$\iint_D f(x, y)\, dA = \iint_R F(x, y)\, dA = \int_a^b \int_c^d F(x, y)\, dy\, dx$$

Observe that $F(x, y) = 0$ if $y < g_1(x)$ or $y > g_2(x)$ because (x, y) then lies outside D. Therefore

$$\int_c^d F(x, y)\, dy = \int_{g_1(x)}^{g_2(x)} F(x, y)\, dy = \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy$$

because $F(x, y) = f(x, y)$ when $g_1(x) \le y \le g_2(x)$.

General Regions (10 of 12)

Thus we have the following formula that enables us to evaluate the double integral as an iterated integral.

3 If f is continuous on a type I region D described by

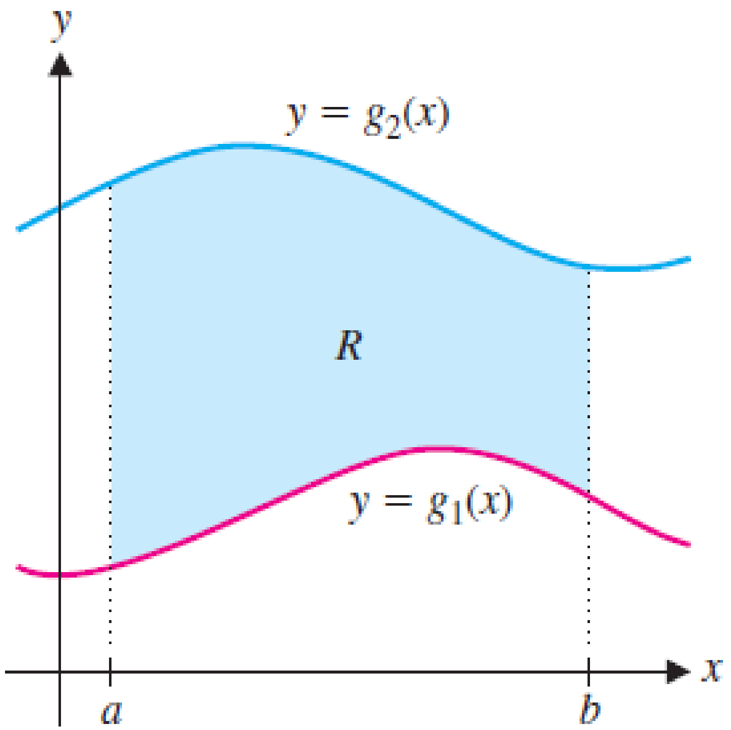
$$D = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$$

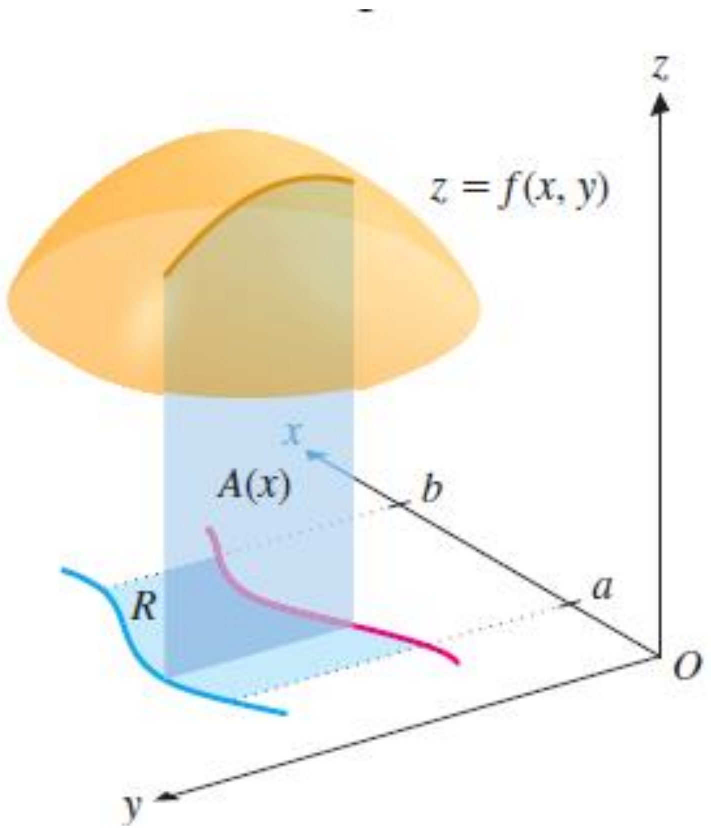
then

$$\iint_D f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$

The integral on the right side of (3) is an iterated integral, except that in the inner integral we regard x as being constant not only in $f(x, y)$ but also in the limits of integration, $g_1(x)$ and $g_2(x)$.

$$R = \{(x, y) | a \le x \le b \text{ and } g_1(x) \le y \le g_2(x)\}.$$





$$A(x) = \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy.$$

$$V = \int_a^b A(x)\, dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$

Example 1

Evaluate $\iint_D (x + 2y)\, dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

Solution:

The parabolas intersect when $2x^2 = 1 + x^2$, that is, $x^2 = 1$, so $x = \pm 1$.

We note that the region D, sketched in Figure 8, is a type I region but not a type II region and we can write

$$D = \{(x,\ y) \mid -1 \le x \le 1,\ 2x^2 \le y \le 1 + x^2\}$$

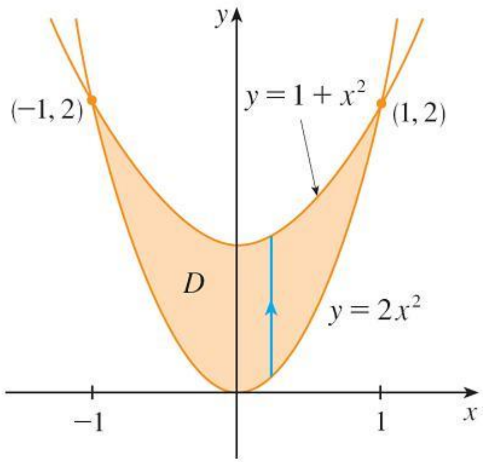


Figure 8

Example 1 – Solution (1 of 2)

Since the lower boundary is $y = 2x^2$ and the upper boundary is $y = 1 + x^2$, Equation 3 gives

$$\iint_D (x + 2y)\, dA = \int_{-1}^{1} \int_{2x^2}^{1+x^2} (x + 2y)\, dy\, dx$$

$$= \int_{-1}^{1} \Big[xy + y^2 \Big]_{y=2x^2}^{y=1+x^2} dx$$

$$= \int_{-1}^{1} [x(1 + x^2) + (1 + x^2)^2 - x(2x^2) - (2x^2)^2]\, dx$$

Example 1 – Solution (2 of 2)

$$= \int_{-1}^{1} (-3x^4 - x^3 + 2x^2 + x + 1)\, dx$$

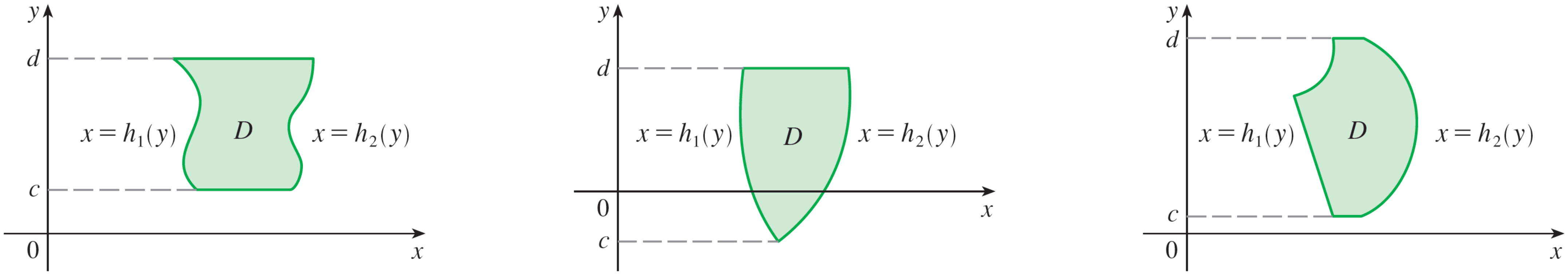
$$= -3\frac{x^5}{5} - \frac{x^4}{4} + 2\frac{x^3}{3} + \frac{x^2}{2} + x \Big]_{-1}^{1}$$

$$= \frac{32}{15}$$

We also consider plane regions of **type II**, which can be expressed as

$$D = \{(x,\ y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$$

where h_1 and h_2 are continuous. Two such regions are illustrated in Figure 7.



Some type II regions

Figure 7

General Regions (12 of 12)

Using the same methods that were used in establishing (3), we can show that the following result holds.

4 If *f* is continuous on a type II region *D* described by

$$D = \{(x,\ y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$$

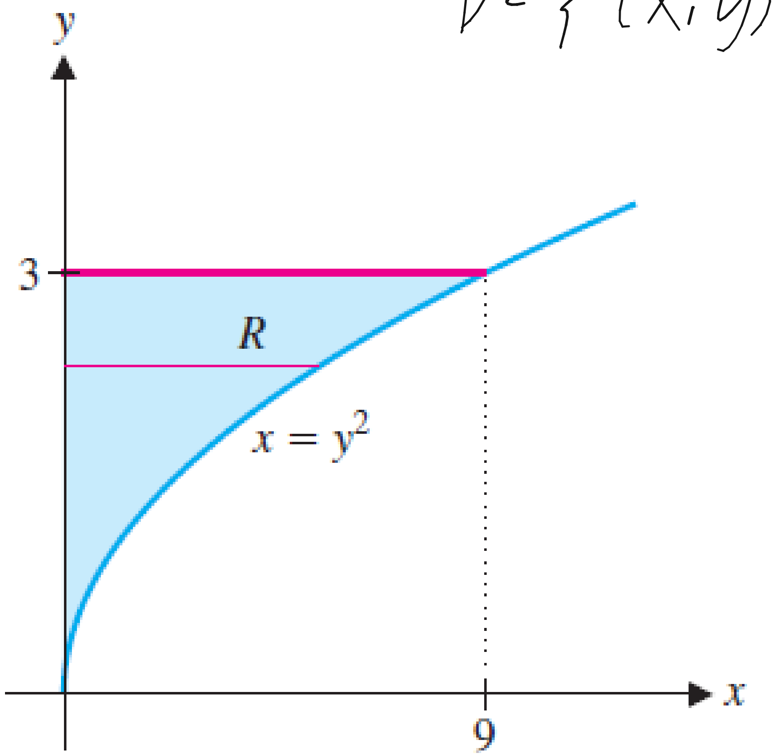
then

$$\iint_D f(x,\ y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,\ y)\, dx\, dy$$

EXAMPLE 1.6 Evaluating a Double Integral

Let R be the region bounded by the graphs of $y = \sqrt{x}$, $x = 0$ and $y = 3$. Evaluate $\iint\limits_R (2xy^2 + 2y \cos x)\, dA$.

$D = \{(x, y) \mid 0 \le x \le 9,\ 0 \le y \le 3\}$



Changing the Order of Integration

Changing the Order of Integration (1 of 1)

Fubini's Theorem tells us that we can express a double integral as an iterated integral in two different orders. Sometimes one order is much more difficult to evaluate than the other—or even impossible. The next example shows how we can change the order of integration when presented with an iterated integral that is difficult to evaluate.

Example 5

Evaluate the iterated integral $\int_0^1 \int_x^1 \sin(y^2)\,dy\,dx$.

Solution:

If we try to evaluate the integral as it stands, we are faced with the task of first evaluating $\int \sin(y^2)\,dy$. But it's impossible to do so in finite terms since $\int \sin(y^2)\,dy$ is not an elementary function. So we must change the order of integration. This is accomplished by first expressing the given iterated integral as a double integral. Using (3) backward, we have

$$\int_0^1 \int_x^1 \sin(y^2)\,dy\,dx = \iint_D \sin(y^2)\,dA$$

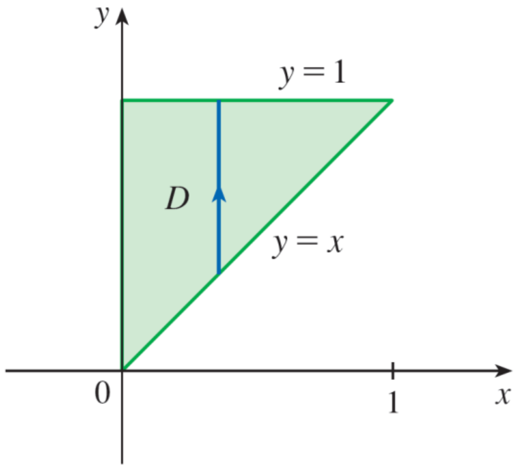
where

$$D = \{(x, y) \mid 0 \le x \le 1,\ x \le y \le 1\}$$

Example 5 – Solution (1 of 2)

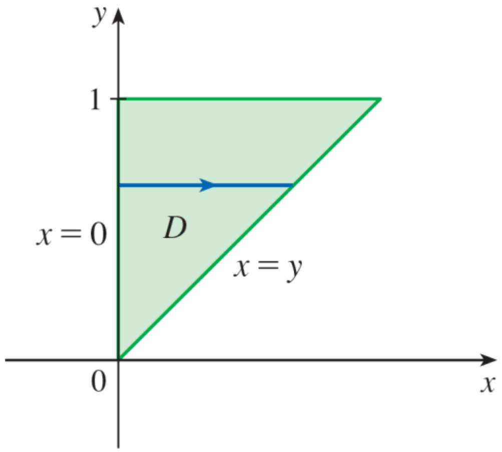
We sketch this region *D* in Figure 15. Then from Figure 16 we see that an alternative description of *D* is

$$D = \left\{(x,\ y) \middle| 0 \le y \le 1,\ 0 \le x \le y\right\}$$



D as a type I region

Figure 15



D as a type II region

Figure 16

Example 5 – Solution (2 of 2)

This enables us to use (4) to express the double integral as an iterated integral in the reverse order:

$$\int_0^1 \int_x^1 \sin(y^2)\,dy\;dx = \iint_D \sin(y^2)\,dA$$

$$= \int_0^1 \int_0^y \sin(y^2)\,dx\;dy = \int_0^1 \Big[x \sin(y^2)\Big]_{x=0}^{x=y} dy$$

$$= \int_0^1 y \sin(y^2)\,dy = -\frac{1}{2}\cos(y^2)\Big]_0^1 = \frac{1}{2}(1 - \cos 1)$$

Properties of Double Integrals

Properties of Double Integrals (1 of 6)

We assume that all of the following integrals exist. For rectangular regions *D* the first three properties can be proved in the same manner. And then for general regions the properties follow from Definition 2.

5 $$\iint_D [f(x, y) + g(x, y)]\, dA = \iint_D f(x, y)\, dA + \iint_D g(x, y)\, dA$$

6 $$\iint_D cf(x, y)\, dA = c\iint_D f(x, y)\, dA \qquad \text{where } c \text{ is a constant}$$

If $f(x, y) \geq g(x, y)$ for all (x, y) in D, then

7 $$\iint_D f(x, y)\, dA \geq \iint_D g(x, y)\, dA$$

Properties of Double Integrals (2 of 6)

The next property of double integrals is similar to the property of single integrals given by the equation

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

If $D = D_1 \cup D_2$, where D_1 and D_2 don't overlap except perhaps on their boundaries (see Figure 17), then

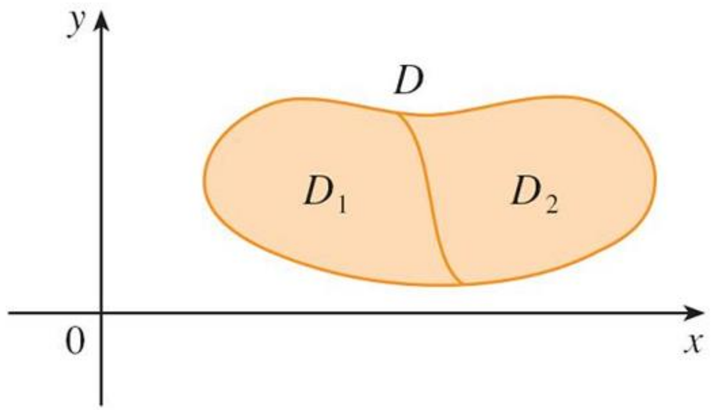
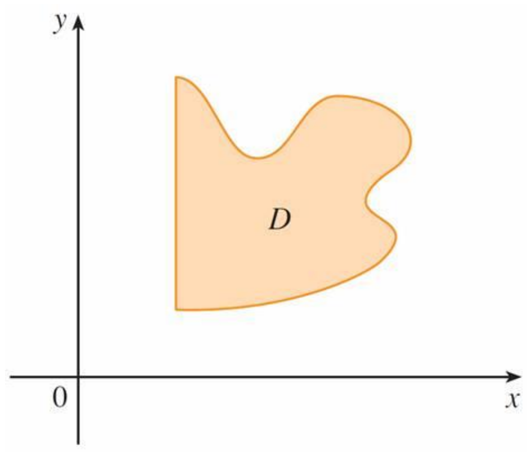


Figure 17

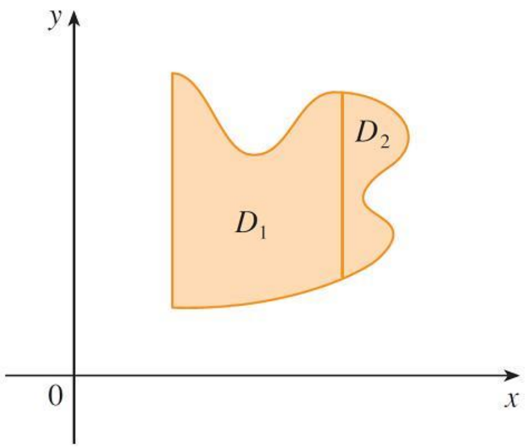
8 $$\iint_D f(x, y)\,dA = \iint_{D_1} f(x, y)\,dA + \iint_{D_2} f(x, y)\,dA$$

Properties of Double Integrals (3 of 6)

Property 8 can be used to evaluate double integrals over regions *D* that are neither type I nor type II but can be expressed as a union of regions of type I or type II. Figure 18 illustrates this procedure.



(a) D is neither type I nor type II.



(b) $D = D_1 \cup D_2$, D_1 is type I, D_2 is type II.

Figure 18

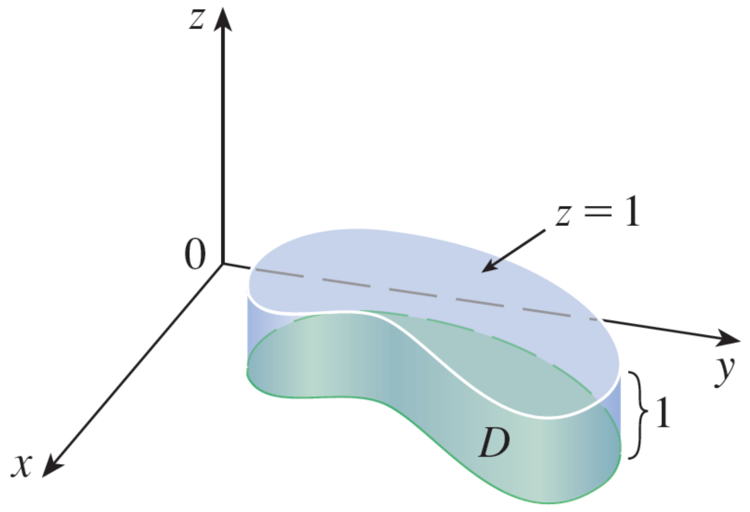
Properties of Double Integrals (4 of 6)

The next property of integrals says that if we integrate the constant function $f(x, y) = 1$ over a region D, we get the area of D:

$$\mathbf{9} \qquad \iint_D 1 \, dA = A(D)$$

Properties of Double Integrals (5 of 6)

Figure 19 illustrates why Equation 9 is true: A solid cylinder whose base is D and whose height is 1 has volume $A(D) \cdot 1 = A(D)$, but we know that we can also write its volume as $\iint_D 1\, dA$.



Cylinder with base D and height 1

Figure 19

Properties of Double Integrals (6 of 6)

Finally, we can combine Properties 6, 7, and 9 to prove the following property.

10 If $m \le f(x, y) \le M$ for all (x, y) in D, then

$$m \cdot A(D) \le \iint_D f(x, y)\, dA \le M \cdot A(D)$$

15.9 Change of Variables in Multiple Integrals

Change of Variables in Multiple Integrals (1 of 1)

In one-dimensional calculus we often use a change of variable (a substitution) to simplify an integral. By reversing the roles of x and u, we can write

1 $$\int_a^b f(x)\,dx = \int_c^d f(g(u))\,g'(u)\,du$$

where $x = g(u)$ and $a = g(c)$, $b = g(d)$. Another way of writing Formula 1 is as follows:

2 $$\int_a^b f(x)\,dx = \int_c^d f(x(u))\frac{dx}{du}\,du$$

Change of Variables in Double Integrals

Change of Variables in Multiple Integrals (2 of 18)

Consider a change of variables that is given by a **transformation** *T* from the *uv*-plane to the *xy*-plane:

$$T(u, v) = (x, y)$$

where *x* and *y* are related to *u* and *v* by the equations

$$x = g(u, v) \qquad y = h(u, v) \tag{3}$$

or, as we sometimes write,

$$x = x(u, v) \qquad y = y(u, v)$$

Change of Variables in Multiple Integrals (3 of 18)

We usually assume that T is a $\mathbf{C^1}$ **transformation**, which means that g and h have continuous first-order partial derivatives.

A transformation T is really just a function whose domain and range are both subsets of $\mathbb{R}^2$.

If $T(u_1, v_1) = (x_1, y_1)$, then the point (x_1, y_1) is called the **image** of the point (u_1, v_1).

If no two points have the same image, T is called **one-to-one**.

Change of Variables in Multiple Integrals (4 of 18)

Figure 1 shows the effect of a transformation *T* on a region *S* in the *uv*-plane.

T transforms *S* into a region *R* in the *xy*-plane called the **image of *S***, consisting of the images of all points in *S*.

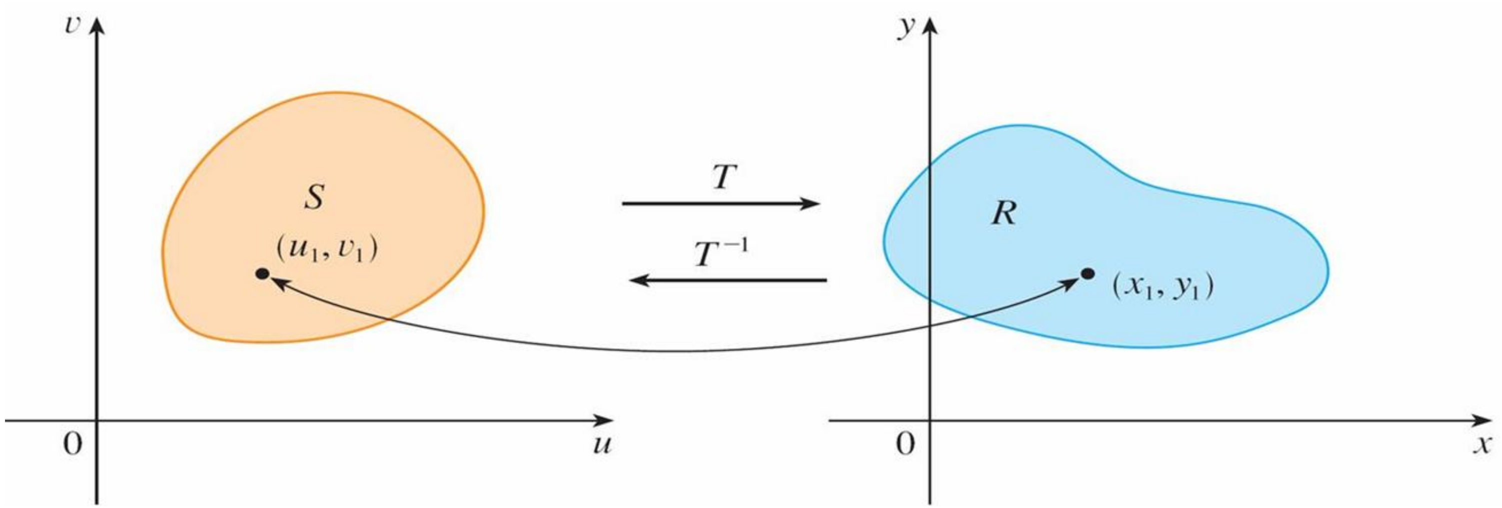


Figure 1

Change of Variables in Multiple Integrals (5 of 18)

If T is a one-to-one transformation, then it has an **inverse transformation** T^{-1} from the xy-plane to the uv-plane and it may be possible to solve Equations 3 for u and v in terms of x and y:

$$u = G(x, y) \qquad v = H(x, y)$$

Change of Variables in Multiple Integrals (12 of 18)

The determinant that arises in this calculation is called the *Jacobian* of the transformation and is given a special notation.

7 Definition The **Jacobian** of the transformation T given by $x = g(u, v)$ and $y = h(u, v)$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

With this notation we can use Equation 6 to give an approximation to the area ΔA of R:

$$\textbf{8} \quad \Delta A \approx \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \Delta u \, \Delta v$$

where the Jacobian is evaluated at (u_0, v_0).

Change of Variables in Multiple Integrals (13 of 18)

Next we divide a region S in the uv-plane into rectangles S_{ij} and call their images in the xy-plane R_{ij}. (See Figure 6.)

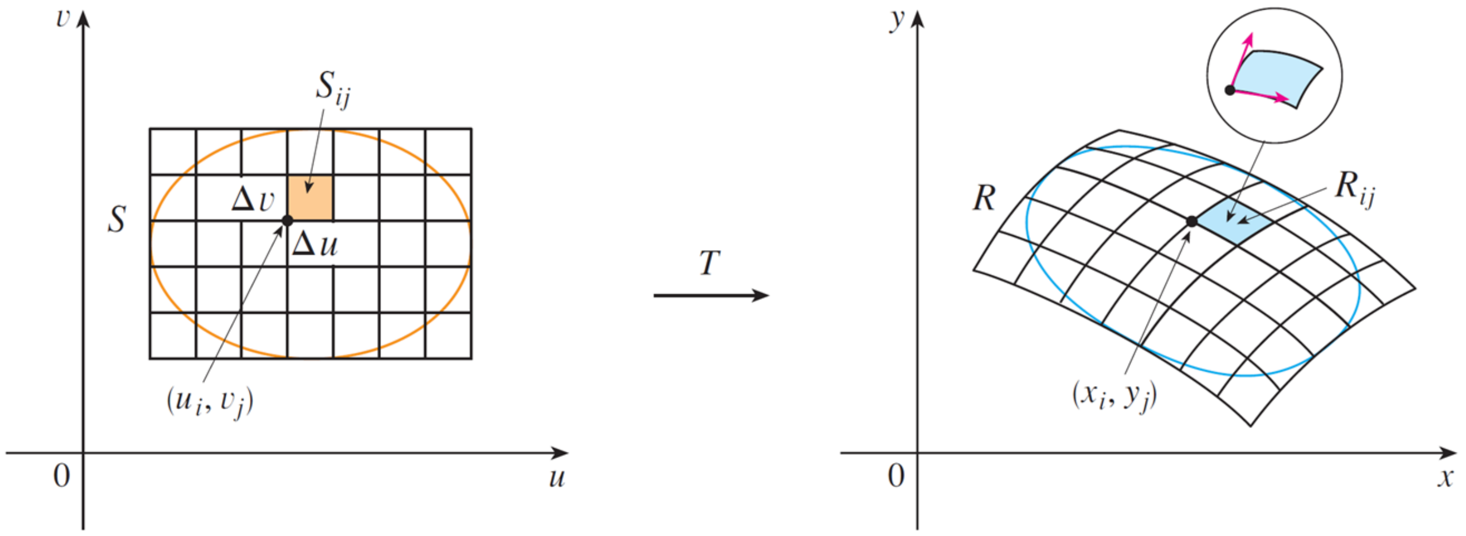


Figure 6

Change of Variables in Multiple Integrals (14 of 18)

Applying the approximation (8) to each R_{ij}, we approximate the double integral of f over R as follows:

$$\iint_R f(x, y)\,dA \approx \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_i, y_i)\,\Delta A$$

$$\approx \sum_{i=1}^{m}\sum_{j=1}^{n} f\left(g(u_i, v_j), h(u_i, v_i)\right)\left|\frac{\partial(x, y)}{\partial(u, v)}\right|\Delta u\,\Delta v$$

where the Jacobian is evaluated at (u_i, v_j). Notice that this double sum is a Riemann sum for the integral

$$\iint_S f\left(g(u, v), h(u, v)\right)\left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\,dv$$

Change of Variables in Multiple Integrals (15 of 18)

The foregoing argument suggests that the following theorem is true.

9 Change of Variables in a Double Integral Suppose that T is a C^1 transformation whose Jacobian is nonzero and that T maps a region S in the uv-plane onto a region R in the xy-plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose also that T is one-to-one, except perhaps on the boundary of S. Then

$$\iint_R f(x, y)\,dA = \iint_S f\big(x(u, v), y(u, v)\big)\left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv$$

Theorem 9 says that we change from an integral in x and y to an integral in u and v by expressing x and y in terms of u and v and writing

$$dA = \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv$$

Change of Variables in Multiple Integrals (16 of 18)

Notice the similarity between Theorem 9 and the one-dimensional formula in Equation 2.

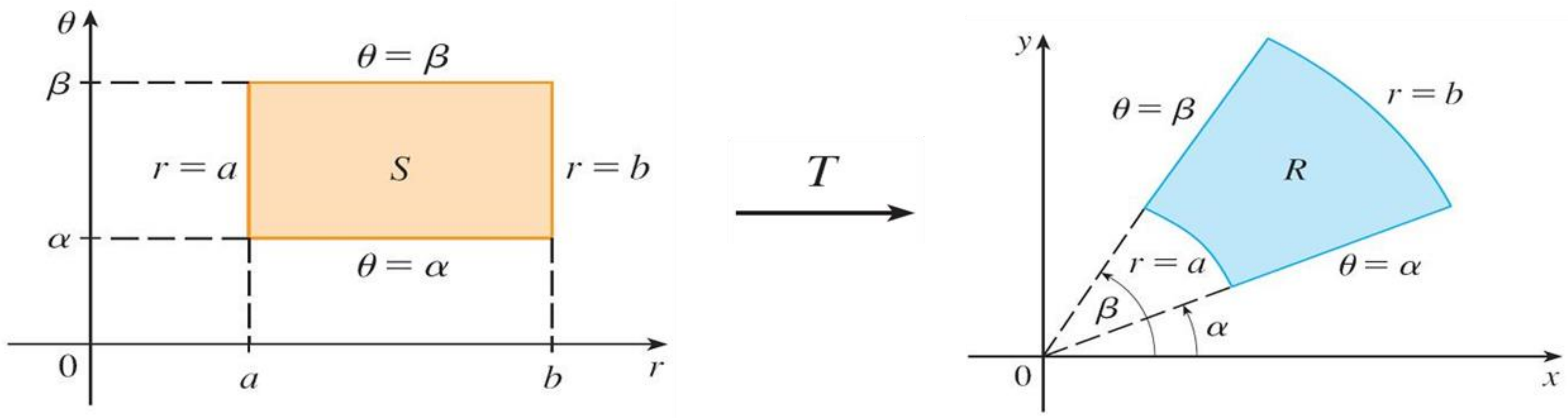
Instead of the derivative $\frac{dx}{du}$, we have the absolute value of the Jacobian, that is, $\left|\frac{\partial(x, y)}{\partial(u, v)}\right|$.

As a first illustration of Theorem 9, we show that the formula for integration in polar coordinates is just a special case.

Here the transformation T from the $r\theta$-plane to the xy-plane is given by

$$x = g(r,\ \theta) = r\ \cos\theta \quad y = h(r,\ \theta) = r\ \sin\theta$$

and the geometry of the transformation is shown in Figure 7. T maps an ordinary rectangle in the $r\theta$-plane to a polar rectangle in the xy-plane.



The polar coordinate transformation

Figure 7

The Jacobian of T is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r > 0$$

Thus Theorem 9 gives

$$\begin{aligned} \iint_R f(x, y)\,dx\,dy &= \iint_S f(r\cos\theta, r\sin\theta)\left|\frac{\partial(x, y)}{\partial(r, \theta)}\right| dr\,d\theta \\ &= \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta \end{aligned}$$